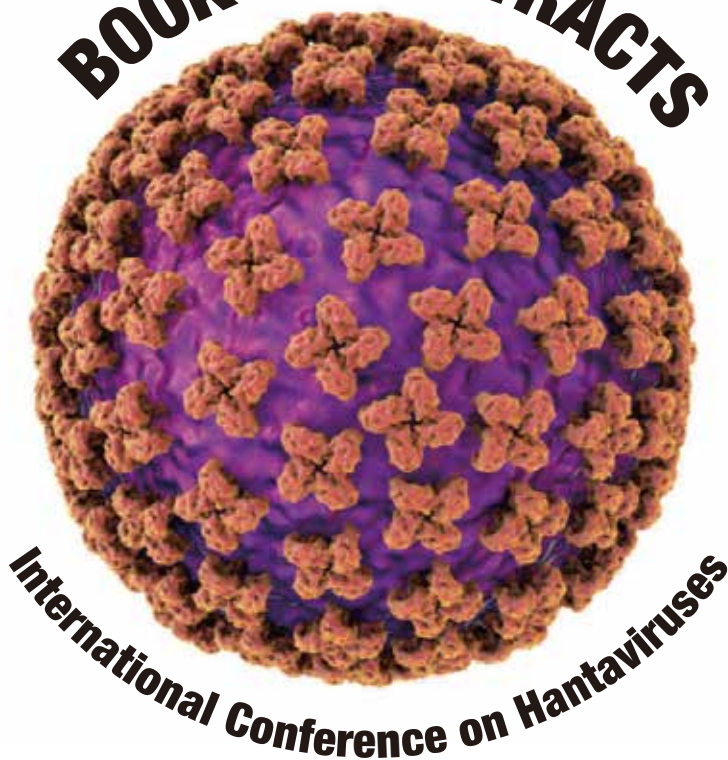


The 12th

ICH 2023

BOOK OF ABSTRACTS



14-16 June 2023

Korea University College of Medicine

Seoul, Korea



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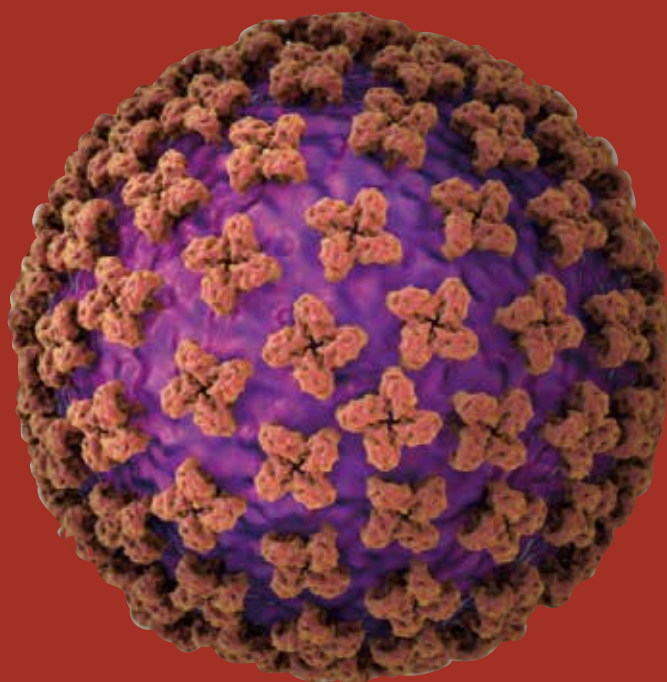
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The Korean Society of Virology



International Society
for Hantaviruses



ICH 2023



BOOK OF ABSTRACTS

12TH INTERNATIONAL CONFERENCE ON HANTAVIRUSES

14-16 June 2023
Seoul, Korea

ICH 2023, 12th International Conference on Hantaviruses, 14-16 June 2023, Seoul

is organized by the Institute for Viral Disease, Korea University College of Medicine, Seoul, Korea.

Book of Abstracts

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CONTENTS

Acknowledgements	4
Organization	5
Local Organizing Committee	
Scientific Committee	
International Advisory Board	
Conference venue	6
Previous awardees	7
History of “International Conference on Hantaviruses”	8
Scientific program	9 - 13
Abstract index	14 - 18
Abstracts	
Awards	19 – 20
Keynote Talks	21 – 26
Oral Presentations	27 – 51
Session 1. Viral epidemiology, evolution & genetics	
Session 2. Replication, Transcription, Translation & Assembly	
Session 3. Immune responses & Pathogenesis	
Session 4. Vaccines, Therapeutics & Clinical aspects	
Session 5. Diagnosis & Emerging Hantaviruses	
Poster Presentations	52 – 90
Session 1. Viral epidemiology, evolution & genetics	
Session 2. Replication, Transcription, Translation & Assembly	
Session 3. Immune responses & Pathogenesis	
Session 4. Vaccines, Therapeutics & Clinical aspects	
Session 5. Diagnosis & Emerging Hantaviruses	
Author index	91

ACKNOWLEDGEMENTS



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Local Organizing Committee

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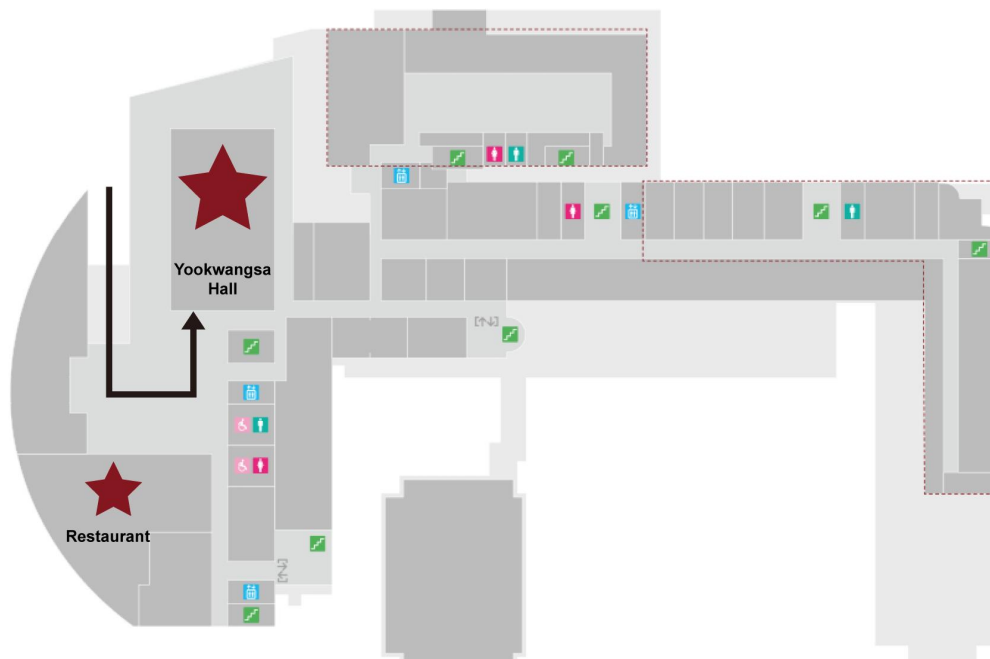
CONFERENCE VENUE

Yookwangsa Hall, Korea University College of Medicine, Seoul, Korea



Korea University College of Medicine,
73, Goryeodae-ro Seong buk-gu,
Seoul 02841, Korea

Floor plan



PREVIOUS AWARDEES

Joel M. Dalrymple Award

- 1995. Finland, Helsinki – Ho Wang Lee
- 1998. Atlanta, GA, U.S. – CJ Peters
- 2001. Annecy, France – Charlie Calisher
- 2004. Seoul, Korea – Juhani Lahdevirta
- 2007. Buenos Aires, Argentina – Connie Schmaljohn
- 2010. Athens, Greece – Jiro Arikawa
- 2013. Beijing, China – Tatjana Avšič-Županc / Richard Yanagihara
- 2016. Fort Collins, CO, U.S. – Irina Gavrilovskaya
- 2019. Leuven, Belgium – Boris Klempa

Ho Wang Lee Award

- 2013. Beijing, China – Jin-Won Song
- 2016. Fort Collins, CO, U.S. – James LeDuc
- 2019. Leuven, Belgium – Antti Vaheri

History of

“International Conference on Hantaviruses”

	Year	Location	Name of Conference
I	1989	Seoul, Republic of Korea	International Conference on HFRS
II	1992	Beijing, China	"
III	1995	Helsinki, Finland	"
IV	1998	Atlanta, U.S.A.	"
V	2001	Annecy, France	International Conference on HFRS, HPS & Hantaviruses
VI	2004	Seoul, Republic of Korea	"
VII	2007	Buenos Aires, Argentina	"
VIII	2010	Athens, Greece	"
IX	2013	Beijing, China	"
X	2016	Colorado, U.S.A.	"
XI	2019	Leuven, Belgium	International Conference on Hantaviruses
XII	2023	Seoul, Republic of Korea	"

SCIENTIFIC PROGRAM

Wednesday, 14/Jun/2023

08:00-09:00am Registraion

09:00-09:10am Welcome Remark

ICH President –Jin-Won Song

President of Korea University – Dong Won Kim

President of Korea University Medicine and Executive Vice President for Medical Affairs of Korea University – Eul-Sik Yoon

09:10-09:25am ICH President Lecture

Memorial lecture for Prof. Ho Wang Lee

09:25-09:30am Ho Wang Lee Award Ceremony

09:30-10:05am Ho Wang Lee Award Lecture

The Hantavirology Continuum - Connie Schmaljohn

Presentation of the awardee of the honorary Ho Wang Lee Award plaque.

10:05-10:35am Keynote 1

How Andes Virus Strain Epuyén Spread: The quest to unravel the Determinants of Human-to-Human Transmission during the Largest Hantavirus Pulmonary Syndrome Outbreak - Gustavo Palacios

10:35-11:10am Coffee Break & Poster viewing

11:10-11:55am Oral Presentation 1-1

Session Chair: Dr. Boris Klempa

11:10-11:25am Genetic Diversity and Phylogeography of Asama Virus in the Japanese Shrew Mole (*Urotrichus talpoides*) - Satoru Arai

11:25-11:40am Hantavirus Prevalence and Viral Strain Distribution in the U.S. 2008-2020 - Shannon Whitmer

11:40-11:55am Hemorrhagic Fever with Renal Syndrome in Russia, 2000-2022 - Evgeniy Tkachenko

11:55-14:30pm Lunch & Poster viewing

14:30-15:15pm Oral Presentation 1-2

Session Chair: Dr. Boris Klempa

14:30-14:45pm Transmission, host distribution, and circulation of Sin Nombre virus at a site of human infection - [Steven B. Bradfute](#)

14:45-15:00pm Flying hantaviruses: What's new in the field of bat-borne hantaviruses?
- [Boris Klempa](#)

15:00-15:15pm The order Bunyavirales and its family Hantaviridae: summary of past and projected taxonomic reorganizations and possible solutions to family-specific virus classification and nomenclature challenges - [Jens H. Kuhn](#)

15:15-15:45pm Keynote 2

Structural and mechanistic basis of neutralization by a pan-hantavirus protective human antibody – [Eva Mittler](#)

15:45-16:15pm Coffee Break & Poster viewing

16:15-17:00pm Oral Presentation 2

Session Chair: Dr. Christina F. Spiropoulou

16:15-16:30pm Conformational Transitions of the Hantavirus Gn/Gc Spikes in Cell-to-Cell Transmission and Entry - [Nicole D. Tischler](#)

16:30-16:45pm Development of a Sensitive and Quantitative Method for Detection of Hantavirus Entry Using NanoLuc Luciferase Complementation Assay
- [Ezgi Kasikci](#)

16:45-17:00pm Role of inhibitory receptors in the pathogenesis of Hemorrhagic Fever with Renal Syndrome - [Željka M. Šafranko](#)

17:30-19:00pm Welcome Dinner

Location: Restaurant

Thursday, 15/Jun/2023

09:25-09:30am J. Dalrymple Award Ceremony

09:30-10:05am J. Dalrymple Award Lecture

Epidemiology, virology and clinical aspects of orthohantavirus infections
- [Misa Korva](#)

Presentation of the awardee of the honorary J. Dalrymple Award plaque.

10:05-10:35am Keynote 3

Hantavirus-infection lead to massive immune activation - but what is good and what is bad? – [Jonas Klingstrom](#)

10:35-11:10am Coffee Break & Poster viewing

11:10-11:55am Oral Presentation 3-1

Session Chair: Dr. Hiroaki Kariwa

11:10-11:25am Absence of Innate Immune Antagonism by Seoul Virus in Reservoir Endothelial Cells - [Alison M Kell](#)

11:25-11:40am Andes virus variants generated in vitro are attenuated in Syrian Golden hamsters because they are unable to evade the innate immune response - [Joseph W. Golden](#)

11:40-11:55am A novel three-dimensional vessel-on-chip model to study orthohantavirus pathogenesis - [Danny Noack](#)

11:55-14:00pm Lunch & Poster viewing

14:00-14:45pm Oral Presentation 3-2

Session Chair: Dr. Hiroaki Kariwa

14:00-14:15pm A human lung xenograft mouse model of pan-orthohantavirus infection - [Danny Noack](#)

14:15-14:30pm A pro-inflammatory IgA response in acute Puumala Orthohantavirus-caused Hemorrhagic Fever with Renal Syndrome - [Tomas Strandin](#)

14:30-14:45pm Two point mutations in protocadherin-1 afford protection against a lethal infection by Andes hantavirus - [Stephanie Monticelli](#)

14:45-15:15pm Keynote 4

Hantavirus Nucleic Acid Vaccines and Their Use in Antibody-based Medical Countermeasure Development – [Jay W. Hooper](#)

15:15-15:45pm Coffee Break & Poster viewing

15:45-17:15pm Oral Presentation 4

Session Chair: Dr. Piet Maes

- 15:45-16:00pm High-Throughput Drug Repurposing Screen for Antivirals Targeting Orthohantaviruses - Wanda Christ
- 16:00-16:15pm Neutralizing Antibody Titers in Hospitalized Patients with Acute Puumala Orthohantavirus Infection Do Not Associate with Disease Severity - R Iheozor-Ejiofor
- 16:15-16:30pm A Novel Disease Model for Sin Nombre Virus Infection in STAT2 Knockout Syrian Golden Hamsters - Catalina Florez
- 16:30-16:45pm Validation of an antigenic site targeted by monoclonal antibodies against Puumala virus - Lev Levanov
- 16:45-17:00pm Persistence and long-term efficacy of neutralizing antibody treatment against Puumala orthohantavirus in the bank vole infection model - Janne Tynell
- 17:00-17:15pm Clinical and laboratory parameters of hantavirus patients in the Czech Republic - Hana Zelena

17:30-20:00pm Conference Dinner

Location: Sudang Samyang faculty house (Paid person only)

Friday, 16/Jun/2023

09:20-09:30am (Next) ICH President Announcement

09:30-10:00am Keynote 5

Using next-generation sequencing to reveal genetic diversity and evolution of Paraguayan hantaviruses in rodent reservoirs in the Neotropics
– Colleen B. Jonsson

10:00-10:30am Oral Presentation 5-1

Session Chair: Dr. Steven B. Bradfute

- 10:00-10:15am The Identification of A Novel Etiological Agent of Hemorrhagic Fever with Renal Syndrome in Southern Areas, The Republic of Korea Using Multiplex PCR-based MinION Sequencing - Sara P Prayitno
- 10:15-10:30am Identification and Complete Coding Sequence of Limestone Canyon virus in Wild-Caught Peromyscus boylii from New Mexico, USA - Samuel Goodfellow

10:30-11:10am Coffee Break & Poster viewing

11:10-11:40am Oral Presentation 5-2

Session Chair: Dr. Steven B. Bradfute

11:10-11:25am Urinary Genome Detection and Surveillance of Hantaan Orthohantavirus from Hemorrhagic Fever with Renal Syndrome Patients Using Amplicon-based Next-generation Sequencing - Won-Keun Kim

11:25-11:40am Use of a New World hantavirus RT-qPCR assay to detect multiple hantavirus in the Americas - Maria Morales-Betoulle

11:40-12:00pm Closing Remarks

ABSTRACT INDEX

Awards

- **Ho Wang Lee Award Ceremony and Lecture**
 - **Dr. Connie Schmaljohn** (Integrated Research Facility, National Institute of Allergy and Infectious Diseases, National Institute of Health, Bethesda, Maryland, USA) 19
"The Hantavirology Continuum"
- **J. Dalrymple Award Ceremony and Lecture**
 - **Dr. Misa Korva** (Institute of Microbiology and Immunology, Faculty of Medicine, University of Ljubljana, Ljubljana, Slovenia) 20
"Epidemiology, virology and clinical aspects of orthohantavirus infections"

Keynote Talks

Session1. Viral epidemiology, evolution & genetics

- Dr. Gustavo Palacios** (Icahn School of Medicine at Mount Sinai, USA)21-22
"How Andes Virus Strain Epuyén Spread: The quest to unravel the Determinants of Human-to-Human Transmission during the Largest Hantavirus Pulmonary Syndrome Outbreak"

Session2. Replication, Transcription, Translation & Assembly

- Dr. Eva Mittler** (Albert Einstein College of Medicine, USA) 23
"Structural and mechanistic basis of neutralization by a pan-hantavirus protective human antibody"

Session3. Immune responses & Pathogenesis

- Dr. Jonas Klingström** (Karolinska Institute, Sweden) 24
"Hantavirus-infection lead to massive immune activation - but what is good and what is bad?"

Session4. Vaccines, Therapeutics & Clinical aspects

- Dr. Jay W. Hooper** (United States Army Medical Research Institute of Infectious Diseases, USA) 25
"Hantavirus Nucleic Acid Vaccines and Their Use in Antibody-based Medical Countermeasure Development"

Session5. Diagnosis & Emerging Hantaviruses

- Dr. Colleen B. Jonsson** (University of Tennessee, USA) 26
"Using next-generation sequencing to reveal genetic diversity and evolution of Paraguayan hantaviruses in rodent reservoirs in the Neotropics"

Oral Presentations

Session1. Viral epidemiology, evolution & genetics

- Genetic Diversity and Phylogeography of Asama Virus in the Japanese Shrew Mole (*Urotrichus talpoides*) - Satoru Arai 27
- Hantavirus Prevalence and Viral Strain Distribution in the U.S. 2008-2020 - Shannon Whitmer 28
- Hemorrhagic Fever with Renal Syndrome in Russia, 2000-2022 - Evgeniy Tkachenko 29
- Transmission, host distribution, and circulation of Sin Nombre virus at a site of human infection - Steven B. Bradfute 30
- Flying hantaviruses: What's new in the field of bat-borne hantaviruses? - Boris Klempa 31
- The order *Bunyavirales* and its family *Hantaviridae*: summary of past and projected taxonomic reorganizations and possible solutions to family-specific virus classification and nomenclature challenges - Jens H. Kuhn 32

Session2. Replication, Transcription, Translation & Assembly

- Conformational Transitions of the Hantavirus Gn/Gc Spikes in Cell-to-Cell Transmission and Entry - Nicole D. Tischler 33
- Development of a Sensitive and Quantitative Method for Detection of Hantavirus Entry Using NanoLuc Luciferase Complementation Assay - Ezgi Kasikci 34
- Role of inhibitory receptors in the pathogenesis of Hemorrhagic Fever with Renal Syndrome - Željka M. Šafranko 35

Session3. Immune responses & Pathogenesis

- Absence of Innate Immune Antagonism by Seoul Virus in Reservoir Endothelial Cells - Alison M Kell 36
- Andes virus variants generated in vitro are attenuated in Syrian Golden hamsters because they are unable to evade the innate immune response - Joseph W. Golden 37
- A novel three-dimensional vessel-on-chip model to study orthohantavirus pathogenesis - Danny Noack 38
- A human lung xenograft mouse model of pan-orthohantavirus infection - Danny Noack 39
- A pro-inflammatory IgA response in acute Puumala Orthohantavirus-caused Hemorrhagic Fever with Renal Syndrome - Tomas Strandin 40
- Two point mutations in protocadherin-1 afford protection against a lethal infection by Andes hantavirus - Stephanie Monticelli 41

Session4. Vaccines, Therapeutics & Clinical aspects

- High-Throughput Drug Repurposing Screen for Antivirals Targeting Orthohantaviruses - Wanda Christ 42
- Neutralizing Antibody Titers in Hospitalized Patients with Acute Puumala Orthohantavirus Infection Do Not Associate with Disease Severity - R Iheozor-Ejiofor 43
- A Novel Disease Model for Sin Nombre Virus Infection in STAT2 Knockout Syrian Golden Hamsters - Catalina Florez 44
- Validation of an antigenic site targeted by monoclonal antibodies against Puumala virus - Lev Levanov 45
- Persistence and long-term efficacy of neutralizing antibody treatment against Puumala orthohantavirus in the bank vole infection model - Janne Tynell 46
- Clinical and laboratory parameters of hantavirus patients in the Czech Republic - Hana Zelena 47

Session5. Diagnosis & Emerging Hantaviruses

- The Identification of A Novel Etiological Agent of Hemorrhagic Fever with Renal Syndrome in Southern Areas, The Republic of Korea Using Multiplex PCR-based MinION Sequencing
- Sara P Prayitno..... 48
- Identification and Complete Coding Sequence of Limestone Canyon virus in Wild-Caught *Peromyscus boylii* from New Mexico, USA - Samuel Goodfellow..... 49
- Urinary Genome Detection and Surveillance of Hantaan Orthohantavirus from Hemorrhagic Fever with Renal Syndrome Patients Using Amplicon-based Next-generation Sequencing
- Won-Keun Kim..... 50
- Use of a New World hantavirus RT-qPCR assay to detect multiple hantavirus in the Americas
- Maria Morales-Betoulle 51

Poster Presentations

Session1. Viral epidemiology, evolution & genetics

- Evidence of orthohantavirus and leptospira infections in small mammals in an endemic area of Gampaha district in Sri Lanka 52
- Hantavirus infection as a risk factor for chronic kidney disease of unknown etiology (CKDu) in Sri Lanka 53
- Widespread presence of Brno loanvirus in common noctule bats (*Nyctalus noctula*) in Central Europe 54
- Phylogeny of Shrew- and Mole-Borne Hantaviruses in Poland and Ukraine..... 55
- Brno loanvirus (BRNV) in bats inhabiting urban area of Brno, Czech Republic..... 56
- A novel genotype of Hantaan virus from *Apodemus chejuensis* from Jeju Island as a potential etiologic agent of hemorrhagic fever with renal syndrome in Republic of Korea..... 57
- A Potential Relationship Between Sin Nombre Hantavirus and the Microbiome of the Deer Mouse, *Peromyscus maniculatus* 58
- Two divergent hantaviruses, Biya River and Boginia, in the Eurasian water shrew (*Neomys fodiens*) in Siberia..... 59
- Development of a Seroepidemiological Tool for Bat-Borne and Shrew-borne Hantaviruses and Its Application for Samples from Zambia 60
- Estimating time-scale of hantavirus evolution 61
- Exploring the Molecular Evolution and Genetic Diversity of Hantaan and Seoul Viruses: A Comparative Genomic Analysis..... 62
- Hantavirus Infection During a Wild Rodent Population Outbreak “Ratada” in Southern Brazil 63
- Is *Carollia perspicillata* a Natural Host of a Novel Hantavirus in the Brazilian Atlantic Rainforest?. 64
- First molecular evidence of shrew-borne Seewis orthohantavirus in Croatia..... 65

Session2. Replication, Transcription, Translation & Assembly

- Molecular Basis for the Host Adaptation of Zoonotic Orthohantaviruses 66
- Analysis of relationship between replication of Hokkaido orthohantavirus and autophagy 67

Session3. Immune responses & Pathogenesis

- Epithelial Cell Proliferation in Kidneys and Relationship With Changes in Acid and Alkaline Phosphatases in Hemorrhagic Fever With Renal Syndrome 68
- Innate Lymphoid Cells are Activated in Hantavirus Infected Patients and can be Activated by Hantavirus Induced Type I Interferons 69
- A study of Puumala virus infection during pregnancy using samples from pregnant women and human trophoblast stem cells 70
- Humoral Immune Characterization of Sin Nombre Hantavirus Survivors 71
- Dobrava and Puumala orthohantaviruses replicate differently in human astrocytes 72
- Defective hantavirus genomes - friend or foe? 73
- Primary human monocytes differentiate into M2-like macrophages upon long-term orthohantaviruses infection 74
- Orthohantavirus reprogramming of transcriptional cellular response in primary human monocytes 75
- Expansion of low-density granulocytes by Puumala orthohantavirus infection 76

Session4. Vaccines, Therapeutics & Clinical aspects

- Development of a novel minigenome and recombinant VSV expressing Seoul hantavirus glycoprotein-based assays to identify anti-hantavirus therapeutics..... 77
- Use of Ribavirin as treatment for Andes South Hantavirus in Chubut, Argentina..... 78
- Clinical and Immunological Biomarkers of Hemorrhagic Fever with Renal Syndrome Outcome during The Early Disease Progression..... 79
- New Approaches for the HFRS-Puumala Vaccine Development 80
- Influence of the Immunization Schedule on the Immune Response to Experimental Inactivated PUUV-Vaccine 81
- Immunomodulatory Effect of Adjuvants in Hantavirus Vaccine in Guinea Pig Model..... 82
- Replication-competent Recombinant Vesicular Stomatitis Viruses Evaluated as Potential Vaccine Candidates Against Old World Hantaviruses 83
- Cardiac arrhythmia and reduced heart rate variability in patients with Puumala orthohantavirus infection..... 84

Session5. Diagnosis & Emerging Hantaviruses

- *Puumala orthohantavirus* diagnostic at the Swedish National Reference Laboratory 85
- Amplicon-based nanopore sequencing and rapid genome surveillance of Hantaan virus from *Apodemus agrarius*, Republic of Korea 86
- A rapid mobile diagnostic assay of *Seoul orthohantavirus* harbored by *Rattus norvegicus* collected in Gangwon Province, Republic of Korea 87
- VirPipe: An Easy-to-use and Customizable Pipeline for Detecting Hantaviral Genomes from Next-generation Sequencing..... 88
- On-site Rapid Detection and Monitoring of Hantaan orthohantavirus Using Real-time PCR in Wild Rodents (*Apodemus agrarius*) 89
- Development and Evaluation of a rapid molecular diagnostic system as duplex qRT-PCR for differentiation of Hantaan/Seoul viruses in single slot 90

ABSTRACTS

Awards

- Ho Wang Lee Award Ceremony and Lecture

The Hantavirology Continuum

Connie S. Schmaljohn, Ph.D.

Director, Integrated Research Facility at Fort Detrick, Division of Clinical Research, National Institute of Allergy and Infectious Diseases, National Institutes of Health, 8200 Research Plaza, Fort Detrick, Frederick, MD 21702, USA

The hantavirology continuum spans a mere four decades. Hantavirology began in the late 1970s when Professor Ho Wang Lee discovered that Korean hemorrhagic fever patient sera reacted with antigen in the lungs of *Apodemus agrarius* mice. Since then, Ho Wang Lee and hantavirology have been inseparably intertwined. I became involved in the study of hantaviruses in the early 1980s when I demonstrated that the newly isolated "Korean hemorrhagic fever virus" had three genome segments of single-strand negative sense RNA; thus, was most likely a member of the family *Bunyaviridae*. Over the next decade, rapid but incremental discoveries about the newly named "Hantaan virus", its' serological relatives, and the disease syndrome that they cause ensued. The discovery in the early 1990s of hantaviruses as the source of an even more deadly disease syndrome provided an impetus for new research on hantaviruses. Today, we are still discovering novel hantaviruses in a myriad of hosts all over the world. The Ho Wang Lee lectureship award honors the legacy of the founder of hantavirology and the International Society of Hantaviruses. With the passing of Professor Ho Wang Lee less than one year ago, the 12th International Conference on Hantaviruses has a special significance. The future of hantavirology and of this society are now in the hands of those of us who follow and those who will follow us. It is a great honor for me to present a treatise on the hantavirology continuum in honor of Professor Lee.

- J. Dalrymple Award Ceremony and Lecture

Epidemiology, virology and clinical aspects of orthohantavirus infections

Miša Korva

Institute of Microbiology and Immunology, Faculty of Medicine, University of Ljubljana

The complex ecology of the Balkan Peninsula supports the existence of several rodent and insectivore species that harbour several pathogenic and nonpathogenic orthohantaviruses. The incidence of haemorrhagic fever with renal syndrome (HFRS) in the Balkans is modest, with about a hundred reported cases in most years, but the frequency of outbreaks has increased in recent years. Within geographic regions, there are certain areas where hantavirus infections have been confirmed in rodents and where human infections are most prevalent. These areas are mostly associated with major water bodies in Slovenia, most often the major rivers. Although both Dobrava (DOBV) and Puumala (PUUV) orthohantaviruses cause disease in humans, only PUUV causes epidemics where the local incidence can reach more than 200/100,000. The development of a sensitive molecular approach allowed us not only to detect early cases but also to study the impact of viral kinetics on disease progression. We have shown that DOBV-infected patients have much higher and longer viremia than PUUV-infected patients. Using our HFRS disease severity scale, we showed that PUUV infections caused mild to moderate clinical manifestations in 89% of patients, whereas very severe disease was recorded in 3% of patients. In patients with DOBV infections, mild disease was seen in 44% and moderate-to-severe disease in 56% of patients, with an overall mortality rate of 9.8%. The pathogenesis of hantavirus is likely a complex, multifactorial process involving the immune response, platelet dysfunction, and deregulation of endothelial cell barrier functions. Although the humoral immune response to infection is already present when the first symptoms of HFRS appear, DOBV-infected patients have lower IgG antibody responses and also a more rapid decline in IgM antibodies. Elevated levels of IL - 10, TNF- α and INF - γ were seen in all patients regardless of the pathogen. However, in DOBV-infected patients, the change in cytokine secretion occurred around day 20 post infection, whereas it occurred earlier in PUUV-infected patients. Also, our recent studies on the neurotrophic affinity of DOBV and PUUV have shown that DOBV inhibits the activation of the immune response in astrocytes, allowing unlimited virus proliferation and eventually leading to a more severe form of the disease.

Keynote Talks

Session1. Viral epidemiology, evolution & genetics

- **How Andes Virus Strain Epuyén Spread: The quest to unravel the Determinants of Human-to-Human Transmission during the Largest Hantavirus Pulmonary Syndrome Outbreak**

Dr. Gustavo Palacios

Icahn School of Medicine at Mount Sinai, USA

"Andes virus (ANDV) is a major cause of hantavirus pulmonary syndrome (HPS) in South America, particularly in Chile and Argentina. The virus is transmitted to humans through contact with the excreta of infected rodents or through person-to-person transmission. A recent HPS outbreak in Chubut Province, Argentina, resulted in 34 confirmed infections and 11 deaths due to person-to-person transmission of ANDV. The outbreak was driven by three symptomatic individuals who attended crowded social events. Once 18 cases were confirmed, public health officials implemented measures to isolate individuals with confirmed cases and enforce self-quarantine for potential contacts, which likely limited further spread.

The median reproductive number, which represents the number of secondary cases caused by an infected person during the infectious period, was 2.12 before the control measures were enforced. It decreased to 0.96 after the measures were implemented. Specimens from 27 patients were used to perform full genome sequencing of the ANDV strain involved in this outbreak. The strain present (Epuyen/18-19) was found to be similar to the causative strain (Epilink/96) responsible for the first known person-to-person transmission of HPS caused by ANDV, which occurred in El Bolson, Argentina, in 1996.

Clinical investigations of patients with ANDV HPS in this outbreak revealed that individuals with a high viral load and liver injury were more likely to spread the infection compared to others. Disease severity, genomic diversity, age, and hospitalization duration did not show a clear association with secondary transmission. Among patients with ANDV HPS, a higher likelihood of transmission was associated with high viral titers in combination with attendance at large social gatherings or extensive contact among individuals.

To date, no animal models have been described to study orthohantavirus transmission. The lethal Syrian golden hamster model of ANDV infection is an important model for studying the pathogenesis of HPS. In this model, the ANDV rodent strain Chile-9717869 is uniformly lethal and has been extensively used for pathogenesis, vaccination, and therapeutic studies. However, the human CHI-7913 strain, despite its high sequence similarity with Chile-9717869, does not cause lethal disease in Syrian hamsters. CHI-7913 can infect hamsters and replicate to moderate levels but exhibits reduced replication ability in tissues compared to Chile-9717869.

We conducted similar studies using human and rodent strains from the Clade that resulted in person-to-person transmission. Mutations in the ANDV nucleocapsid protein, glycoprotein, and small nonstructural protein open reading frames were found to correlate with mutations typically observed in ANDV strains associated with increased pathogenesis in the small animal model. Finally, three amino acid substitutions, two in the small nonstructural protein and one in the glycoprotein, were only present in the clade of viruses associated with efficient person-to-person transmission.

In conclusion, person-to-person transmission of ANDV has been documented, and a higher likelihood of transmission is associated with high viral titers combined with attendance at large social gatherings or extensive contact among individuals. There are single-nucleotide polymorphisms associated with virulence and transmission that could be used to predict strain-specific ANDV virulence and/or transmissibility. Certain ANDV strains may not cause lethal disease in the classical Syrian hamster model of infection, and further exploration of the differences between lethal and non-lethal strains should provide important insights into the molecular determinants of pathogenic hantavirus infection.”

Session2. Replication, Transcription, Translation & Assembly

● Structural and Mechanistic Basis of Neutralization by a Pan-Hantavirus Protective Human Antibody

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The rodent-borne hantavirus Puumala virus (PUUV) and related agents from the Old World hantavirus clade cause hemorrhagic fever with renal syndrome (HFRS) in humans. Divergent New World hantaviruses, including Andes virus (ANDV), cause a distinct zoonotic disease, hantavirus cardiopulmonary syndrome (HCPS). Although these infections are severe and have substantial case-fatality rates, no FDA-approved hantavirus countermeasures are available. Recent work however suggests that hantavirus disease may be amenable to monoclonal antibody-based therapeutics.

Here, we mined the human antibody response to natural PUUV infection and identified neutralizing antibodies (nAbs) targeting tetrameric Gn/Gc glycoprotein complexes from HFRS-experienced donors. We isolated a single broadly nAb that bound Gn/Gc heterodimers but not Gn alone, strongly suggesting that it recognizes a quaternary epitope encompassing both subunits. Structural studies confirmed this hypothesis and elucidated the nAb's broad activity: it recognizes conserved Gc fusion loop sequences and the main chain of variable Gn sequences, thereby straddling the Gn/Gc heterodimer and locking it in its pre-fusion conformation. Our mechanistic studies demonstrated that the nAb's accelerated dissociation from the divergent ANDV Gn/Gc at endosomal acidic pH limits its potency against this highly lethal virus and corrected this liability through high-throughput antibody engineering yielding an affinity-optimized antibody variant. Single low doses of this optimized nAb protected Syrian hamsters and bank voles challenged with the highly virulent HCPS-causing ANDV and HFRS-causing PUUV, respectively, setting the stage for its development as a pan-hantavirus therapeutic. Moreover, the nAb's mode of viral recognition informs the design and development of broadly protective hantavirus vaccines.

Keywords: Gn/Gc-targeting therapeutic antibody, pan-hantavirus, membrane fusion, antibody engineering

Session3. Immune responses & Pathogenesis

- **Hantavirus-infection lead to massive immune activation - but what is good and what is bad?**

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Our aim is to understand the mechanisms behind hantavirus pathogenesis and the acute and long-term consequences of hantavirus-infection. We focus on the capacity of hantaviruses to affect normal cell signaling and functions, especially cell death, immune responses, and inflammatory responses. The ultimate goal is to generate a better understanding of the details of hantavirus-induced pathogenesis to aid in the development of specific treatment of patients.

FDA/EMA-approved vaccines and specific therapy are lacking. Furthermore, infected patients show increased risks for heart attack and stroke during, and an increased risk for lymphoma after, the acute phase of infection. Altogether, this shows that hantaviruses affect human health in several different ways. In contrast to humans, natural hosts are asymptotically infected - why hantaviruses cause disease in humans is unknown.

Here I will present new and old data regarding systemic immune responses in HFRS- and HPS-patients, and a possible pathogenic mechanism connecting inflammatory responses and increased permeability.

Session4. Vaccines, Therapeutics & Clinical aspects**● Hantavirus Nucleic Acid Vaccines and Their Use in Antibody-based Medical Countermeasure Development**

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Several years ago, we demonstrated that the full-length open reading frame of the hantavirus M segment, optimized and cloned into a plasmid expression vector (DNA vaccine), can produce neutralizing antibodies and protection in animal models. Since that time, we have advanced DNA vaccines targeting Hantaan virus, Puumala virus, and most recently Andes virus into clinical trials. In recent Phase 1 and 2b trials, we have used needle-free, disposable syringe, jet injection technology to deliver the unadjuvanted, unformulated vaccines. The vaccines are safe and immunogenic, including the production of neutralizing antibodies. We have also used these same hantavirus DNA vaccines to produce potent, fully human, polyclonal antibodies in transchromosomal bovines, and demonstrated that these antibodies can protect against both New World and Old World hantaviruses in hamster pre- and postexposure models of disease and/or infection. Lastly, cells transfected with these DNA vaccine plasmids were used to pan for human monoclonal antibodies, and pseudovirion neutralization assays produced using the plasmids were used to measure neutralization activity. One antibody, SNV-53, was shown to be broadly neutralizing and protected against prototype New World and Old World hantaviruses in hamster models. Collectively, these efforts have advanced candidate vaccines and antibodies deep into the proof-of-concept space. Whether any of these approaches results in a licensed medical countermeasure is dependent on finding creative solutions to proving efficacy in humans, and a market capable of supporting development and sustainment of the product.

Session5. Diagnosis & Emerging Hantaviruses

● Using next-generation sequencing to reveal genetic diversity and evolution of Paraguayan hantaviruses in rodent reservoirs in the Neotropics

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Hantaviruses have been discovered in more than fifty species within the Order Rodentia on five continents. In retrospect, this high level of biogeographical diversity of reservoirs should not be surprising given that rodents comprise forty percent of all mammalian species on earth. The geographical distribution of hantaviruses in bats (Order Chiroptera), moles and shrews (Order Soricimorpha) suggests a similar diversity of reservoirs. The explosion of surveillance data has resulted in discussion regarding the evolutionary history of *Mammantavirinae* such as hypotheses for coevolution, cospeciation, and host-shift speciation in mammals. To address these and other critical questions, more studies are needed that address knowledge gaps in wildlife prevalence of hantaviruses, *in vitro* studies of virus-reservoir host interactions, and maintenance of virus populations (inter- and intra-host) in wildlife. These types of studies are challenging as most of the reservoirs, and the viruses they harbor, lack necessary reagents for their study. Moreover, full length genomes and viral isolates are not available for the majority of hantaviruses. Recent advances in next generation sequencing (NGS) and bioinformatic analyses during COVID-19 pandemic, from pathogen discovery to variant characterization illustrate the power of these diagnostic tools that may be adapted for such studies. Recent studies from our laboratory and others have shown the utility of NGS approaches in the study of hantaviruses harbored by wildlife. We will discuss how we have leveraged NGS to probe the genetic diversity of Paraguayan hantaviruses and how these new tools suggest new concepts in inter- and intra-host viral populations within and among rodent reservoirs. Modeling of the molecular clock and phylogeographical spread of hantaviruses within the neotropical forests of Paraguay will be presented.

Oral Presentations

Session1. Viral epidemiology, evolution & genetics

● Genetic Diversity and Phylogeography of Asama Virus in the Japanese Shrew Mole (*Urotrichus talpoides*)

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Recent discovery of genetically distinct hantaviruses in shrews, moles and bats challenges the long-held view that rodents are the principal reservoir hosts and raises the possibility that eulipotyphla and chiroptera may have played an important role in the evolutionary history of hantaviruses. Previously, we identified Asama virus (ASAV) in the Japanese shrew mole (*Urotrichus talpoides*) from Mie Prefecture. The existence of two distinct chromosomal races of Japanese shrew moles, geographically separated by the Fuji and Kurobe rivers in central Honshu, provided an opportunity to study the genetic diversity and phylogeography of ASAV in Japan. Lung or liver tissues from 72 Japanese shrew moles, captured in 14 prefectures (Ehime, Fukushima, Gunma, Ishikawa, Miyazaki, Nara, Niigata, Oita, Okayama, Saitama, Tokyo, Tottori, Wakayama, and Yamanashi) between 1992 and 2020, were analyzed for ASAV RNA by RT-PCR using oligonucleotide primers based on sequences of known hantaviruses. ASAV RNA was detected in Japanese shrew moles from Niigata, Gunma, Okayama and Yamanashi Prefectures. Pair-wise alignment and comparison of partial or entire S-, M- and L-segment sequences showed similarities of 76.1–78.3% and 85.4–90.7% against the prototype ASAV strain N10 at the nucleotide and amino acid levels, respectively. Phylogenetic analyses of the newly acquired sequences, generated by Bayesian methods, indicated topologies suggestive of geographic-specific clustering, similar to the phylogeography of sorcid- and rodent-borne hantaviruses. Host mtDNA sequence analysis indicated two distinct lineages of Japanese shrew moles, congruent with the ASAV genetic diversity. The close genetic and phylogenetic relationship between ASAV strains and hantaviruses harbored by soricine shrews in Eurasia suggests a possible host-switching event prior to the migration of ancestral shrew moles to Japan, hundreds of thousands of years before present.

Key Words; *Hantaviridae*; hantavirus; mole; Japan

● Hantavirus Prevalence and Viral Strain Distribution in the U.S. 2008-2020

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In the U.S., hantavirus pulmonary syndrome (HPS) and non-HPS hantavirus infection are nationally notifiable diseases and here we present an overview of human hantavirus disease prevalence and viral strain distribution in the U.S. from 2008-2020. Criteria for states to report hantavirus-positive human cases is based on clinical symptoms (HPS or non-HPS) and acute laboratory diagnostic support (IgM+, rising IgG+ titers, PCR+ or immunohistochemistry+). Diagnostic testing is performed at U.S. CDC, select state laboratories that run U.S. CDC-specific or commercial assays, and/or commercial laboratories. Historical data acquired at CDC from 2008-2020 (representing 34% of total human hantavirus cases reported by states) demonstrated that individuals with known symptom onset dates were IgM-positive and PCR-positive by 4 days post onset (dpo) and IgG-positive by 21 dpo. Most U.S. hantavirus cases (85%) were detected during acute (IgM+ or IgM+/IgG+) disease, however, some cases were identified in their convalescent phase in states not previously associated with hantavirus infections (Louisiana, Missouri, Tennessee, New Jersey and Massachusetts). All available state-reported data identified that most U.S. hantaviruses cases occur west of the Mississippi with concentrations highest in the Four Corners region of the U.S. and most Sin Nombre virus (SNV) cases occurred seasonally between the months of March-August. Genetic sequencing of positive hantavirus cases demonstrates a geographic separation of SNV sub-strains (SNV, New York virus, and Monongahela virus), and the presence of two SNV subclades, however, there is a large gap in viral sequence data from the Central U.S. Finally, data suggests that commercially available diagnostic assays generate divergent IgM results compared to CDC-specific assays, and that "concordant positive" (i.e. commercial IgM+ and CDC IgM+ results) specimens clinical exhibit characteristics of traditional hantavirus-associated disease (thrombocytopenia, elevated hematocrit, elevated creatinine, IgG+ seroconversion and/or PCR positive results). Using these data, a suggested testing algorithm for hantavirus diagnostics is presented

● Hemorrhagic Fever with Renal Syndrome in Russia, 2000-2022

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During 2000–2022, a total of 69 of Russia's 85 administrative regions reported 164,580 HFRS cases, an annual average rate of 4.9 cases/100,000 (10^5) inhabitants. European Russia reported 162,045 (98.5%) cases in 53/60 regions and 9.7 cases/ 10^5 persons. Asian Russia reported only 2,535 (1.5%) cases in 16/25 regions and 0.6 cases/ 10^5 persons. The highest 140,000 HFRS cases (85% of all Russian HFRS cases) and 20 cases/ 10^5 persons reported in the eastern areas of European Russia.

In 2000–2022, Russia had 668 (0.4%) fatal HFRS cases, 4,030 (2.4%) cases among children <14 years of age. Most HFRS cases in European Russia occurred during the summer and autumn, but cases in Asian Russia – the autumn and winter. The incidence among rural residents 6.7 per 10^5 was higher than urban 4.4 per 10^5 , however, among HFRS patients, rural and urban residents account for 35% and 65%, respectively.

Six hantaviruses, causing HFRS of different clinical severity, have been recognized as pathogens in Russia: HTNV, AMRV, SEOV, PUUV, and two subtypes (KURV and SOCV) of DOBV virus with the principal hosts *Apodemus agrarius mantchuricus*, *Apodemus peninsulae*, *Rattus norvegicus*, *Myodes glareolus*, *Apodemus agrarius agrarius* and *Apodemus ponticus*, respectively. 97.7% of HFRS cases in Russia are reportedly caused by PUUV, therefore this virus plays the main role in the HFRS morbidity structure in Russia.

Keywords: hemorrhagic fever with renal syndrome, zoonotic viral diseases, epidemics, average annual morbidity rate, case-fatality rate

● Transmission, host distribution, and circulation of Sin Nombre virus at a site of human infection

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Sin Nombre orthohantavirus (SNV) is the primary cause of hantavirus cardiopulmonary syndrome in North America. SNV is thought to be carried and transmitted by the western deer mouse, *Peromyscus maniculatus sonoriensis*. Many epidemiologic studies of SNV are based on serology-based surveillance, which is informative but lacks the sensitivity and specificity of SNV genome detection and sequencing. Few reports have used viral genome-based approaches to trace a source of human infection to a given site or to assess circulation of SNV within rodent populations at the same site over time. Here we report an epidemiologic study of a human SNV infection in northern New Mexico, USA, in August of 2020. We conducted rodent trapping at two potential sites of infection—the patient residence and a second site that were separated by ~15 miles (~24 kilometers)—in August 2020, August 2021, and August 2022. SNV genome sequencing from the patient sample and the lungs of trapped rodents identified the patient residence as the likely site of patient infection. Real-time quantitative polymerase chain reaction analysis from the lungs of captured rodents revealed that SNV genetic material was present in a wide range of rodent species in addition to the known host *P. maniculatus sonoriensis*. Ongoing experiments are underway to determine if live SNV can be recovered from these different rodent species. Furthermore, we discovered that the percentage of rodents carrying SNV remained high at the same sites between 2020–2022. These findings highlight the usefulness of hantavirus genome detection and sequencing to identify sites of transmission, discover potential alternate rodent hosts, and demonstrate long-term host transmission in a given geographic region.

Keywords: Sin Nombre, epidemiology, genome, sequencing, host, persistence

● Flying hantaviruses: What's new in the field of bat-borne hantaviruses?

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Bats (order *Chiroptera*) are considered important reservoir hosts for emerging human pathogens. The first bat-borne hantaviruses, Magboi virus from Sierra Leone and Mouyassué virus from Côte d'Ivoire, were discovered as late as 2012. Meanwhile, at least 12 unique bat-borne hantaviruses have been identified in 16 different bat species. Unfortunately, only six of them are currently recognized as virus species due to the limited availability of complete genome sequences. Taxonomically, bat-borne hantaviruses are assigned to the recently established genera *Loanvirus* and *Mobatvirus* of the *Hantaviridae* family. Phylogenetic analyses show that they are not only genetically highly diverse from the rodent-borne hantaviruses (genus *Orthohantavirus*) but, together with shrew- and mole-borne hantaviruses, also evolutionary older. Most recently, a novel hantavirus, named Kiwira virus, was detected in Angolan freetailed bats (*Mops condylurus*) captured in Tanzania and the Democratic Republic of Congo. Phylogenetic analyses indicated Quezon virus from Philippines and Robina virus as its closest relatives. Interestingly, Robina virus is the first and currently the only hantavirus recognized in Australia. It was discovered in a black flying fox (*Pteropus alecto*) sampled near Robina, Queensland, Australia, in 2017, but so far has been published only as a GenBank entry. While the host of Kiwira virus, *M. condylurus*, belongs to the insectivorous free-tailed bats, the putative hosts of Robina virus and Quezon virus belong to the frugivorous flying foxes. The detection of three closely related hantaviruses in taxonomically as well as ecologically diverse bats collected across three continents is quite surprising and further illustrates that the known diversity of bat-borne hantaviruses is vastly incomplete.

Key Words; *Mobatvirus*, evolution, Africa, bat-borne, *Chiroptera*

● **The order *Bunyavirales* and its family *Hantaviridae*: summary of past and projected taxonomic reorganizations and possible solutions to family-specific virus classification and nomenclature challenges**

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The former family *Bunyaviridae* had to be promoted to order *Bunyavirales* in 2017 after the discovery of numerous divergent viruses that were clearly related to, but did not fit into, the then-established five bunyaviral genera *Hantavirus*, *Nairovirus*, *Orthobunyavirus*, *Phlebovirus*, and *Tospovirus*. Since then, these genera were promoted to families with increasingly complex taxonomic suborganizations to accommodate related viruses from often surprisingly divergent hosts. Another megataxonomic reorganization of the order, encompassing a split into two or a promotion to class, is likely imminent due to the very recent discovery of numerous family-rank clades of arthropod, fungal, plant, and protist bunyavirals, some of which even defy the previous dogma of bunyavirals having segmented genomes (e.g., *Tulasviridae*). At the microtaxonomic level, families will have to undergo reorganizations as well—in particular *Hantaviridae* is currently populated by viruses that have not been fully sequenced, thereby likely requiring virus declassifications (distinctly named viruses likely being identical). In addition, the question whether genome segment reassortants constitute novel viruses/represent new species or are “strains” of classified viruses needs to be addressed systematically for all bunyaviral families. Here, I will provide an overview of these developments and challenges, and present possible operational and/or philosophical solutions that will ensure that *Hantaviridae* is reorganized to the best of our current understanding of bunyaviral and hantavirid evolution

Keywords: *Bunyavirales*, classification, *Hantaviridae*, nomenclature, taxonomy.

Session2. Replication, Transcription, Translation & Assembly

● Conformational Transitions of the Hantavirus Gn/Gc Spikes in Cell-to-Cell Transmission and Entry

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All viral fusion protein must establish a balance between the stability of its pre-fusion conformation and the necessary switch to adopt its post-fusion form. At neutral pH, the Gn/Gc spike is highly dynamic at 37°C fluctuating between an open and closed conformation. When fusion is triggered at low pH, the conformational change is induced by ionizable residues. Structural studies have identified multiple critical Gn/Gc contact regions in the hantavirus spikes that include acid-sensitive residues. In this work we have characterized multiple key residues, located in both; Gn and Gc. These residues include Gn His285 next to the η 1 loop. The substitution of His285 into Tyr reduced the Gn/Gc spike stability at neutral pH and lowered the pH required for its fusion activation suggesting its role as protonable switch. We also examined the role of His665 located at the N-terminal region of Gc interacting with domain III, keeping it in its prefusion position. When replacing His665 by Tyr, the fusion activity is lost but allows Gc homotrimerization in which Gc migrates with a larger volume compared to wild type suggesting that the post-fusion positioning of domain III is disrupted. Finally, the Gc Glu757/Asp759 residues form an acid-dependent carboxylate-carboxylic hydrogen bond necessary for structuring of the bc and cd fusion loops. These residues are being analyzed for their membrane partitioning at neutral and acidic pH. As a whole the results suggest a network of acid-sensitive residues necessary for both; the prefusion stability of Gn/Gc spikes and the low pH conformation of Gc.

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● Development of a sensitive and quantitative method for detection of hantavirus entry using NanoLuc luciferase complementation assay

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Hantaviruses are rodent-borne human pathogens associated with two severe diseases: hantavirus cardiopulmonary syndrome and hemorrhagic fever with renal syndrome. There are no FDA-approved antivirals or vaccines for hantavirus at present, and research into viral countermeasures is further restricted by the need to work with most hantaviruses at BSL3 containment. Ongoing efforts have focused on the generation of surrogate systems, which allow studies on hantavirus entry at BSL2. During rescue of replication-competent, recombinant vesicular stomatitis virus (rVSV) bearing Dobrava-Belgrade virus (DOBV) Gn and Gc (Gn/Gc) spike glycoproteins, we identified multiple mutations in the cytoplasmic tail of the Gn subunit. Although these mutations did not impact the infectivity of rVSV-DOBV particles, they significantly increased Gn/Gc's assembly and budding activity to form virus-like particles (VLP). Virus-like particles bearing hantavirus Gn/Gc and packaging a reporter protein could enter cells, affording their use as an alternative to recombinant virus surrogates to study hantavirus entry. We further developed a quantitative entry assay combining the observed increased DOBV Gn/Gc VLP production with a luciferase complementation assay. This assay is based on the high affinity interaction between the split NanoLuc luciferase protein, LgBiT, and the small peptide, HiBiT. We introduced the HiBiT peptide into the Gc C-terminal tail of our 'super-secreting' Gn/Gc mutant and generated VLPs with this protein. Entry of VLP-HiBiT into cells expressing the LgBiT binding partner generated luminescence in a concentration-dependent manner. We anticipate that this approach will fill a gap in the research toolkit by affording entry studies with particles that resemble authentic virions without the need for BSL3 facilities or reporter gene expression assays.

● Role of inhibitory receptors in the pathogenesis of Hemorrhagic Fever with Renal Syndrome

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Puumala orthohantavirus (PUUV) is the leading cause of Hemorrhagic fever with renal syndrome (HFRS) in Croatia. Immune reactions raised against viral antigens are considered hallmarks of HFRS pathogenesis. Inhibitory receptors play a crucial role in the prevention of immunopathogenesis by regulating immune responses, thus preventing tissue damage.

Inhibitory receptor TIM-3 (T-cell immunoglobulin and mucin domain 3) is expressed on exhausted T cells, natural killer (NK) cells, and dendritic cells. It acts as an immune checkpoint receptor that dampens T cell responses. LAG-3 (lymphocyte activation gene 3) is predominantly expressed on activated and regulatory T cells. It negatively regulates T cell activation by inhibiting T cell signaling and cytokine production. PD-1 (programmed cell death protein 1) is widely expressed on activated T and B cells. It serves as a critical regulator of T cell responses and immune tolerance.

In the serum of acute HFRS patients (n=30), TIM-3, LAG-3 and PD-1 protein concentrations were significantly higher compared to healthy controls (n=10). The concentrations of these proteins were heightened in nested PCR-positive serum samples in the acute stage of the disease. In addition, PD-1 protein was elevated in patients with mild compared to moderate and severe clinical pictures.

While in chronic infections, the expression of TIM-3, LAG-3, and PD-1 are found to be detrimental and contribute to viral persistence, our results suggest that in HFRS patients, upregulation of these receptors can have a protective role in the regulation of immunopathogenesis caused by PUUV infection.

Session3. Immune responses & Pathogenesis

● Absence of Innate Immune Antagonism by Seoul Virus in Reservoir Endothelial Cells

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Seoul orthohantavirus (SEOV) is endemic in its reservoir host *Rattus norvegicus*, the brown rat, and has been detected in rat populations in all major global cities, posing a continuous zoonotic threat to humans. We have previously demonstrated that Old World hantavirus infection in human endothelial cells (EC) drives robust antiviral signaling through the cytoplasmic RNA recognition receptors RIG-I and MDA5. We now report that while Hantaan virus (HTNV) infection in primary rat EC also stimulates immune activation, SEOV infection of reservoir cells elicits no antiviral response. Importantly, when RNA from SEOV-infected Vero E6 or rat EC was transfected into rat EC, we measured a significant increase in antiviral gene expression, suggesting that RNA present during infection can trigger immune activation in these cells. *We hypothesized that SEOV deploys mechanisms to antagonize innate immune signaling in rat EC.* To test this, we treated SEOV-infected rat EC with agonists to initiate signaling through RIG-I, MDA5, Toll-like receptor 3, and type I interferon receptor alpha. We observed no change in interferon-stimulated gene expression by immunoblot or RT-PCR when comparing treated cells infected with SEOV with mock-infected controls. When SEOV-infected rat EC were serially infected with HTNV or Sendai virus (known to drive RIG-I activation), we also observed no inhibition of innate immune signaling. We conclude that SEOV does not directly antagonize innate immune signaling cascades, but instead has developed host-specific mechanisms to evade innate immune recognition to facilitate persistence. Future research will investigate the mechanisms of host-specific immune evasion.

Keywords: Seoul virus, innate immunity, antagonism, reservoir, immune evasion

Scientific Session: Immune responses and pathogenesis; viral epidemiology, evolution and genetics

- **Andes virus variants generated in vitro are attenuated in Syrian Golden hamsters because they are unable to evade the innate immune response**

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Hantaviruses are maintained in nature by persistent infection of rodent carrier populations. These long-term interactions are characterized by limited pathogenesis and often continue despite the development of anti-hantavirus adaptive immunity. In contrast, several hantavirus species can cause significant disease in humans. Factors dictating why hantaviruses cause benign infections in rodents yet produce a devastating disease in humans are largely unexplored. There is currently no reverse genetics system for hantaviruses. Through cell culture passaging, we generated Andes virus (ANDV) variants and evaluated their virulence in the Syrian Golden hamster model. Two ANDV variants, p20pp15 and p20pp17, were attenuated in wild-type hamsters as evidenced by limited lung injury, reduced lung viral load, and no mortality. Both nonpathogenic ANDV variants contained an E265K mutation in the glycoprotein, Gn. The attenuated phenotype was lost in cyclophosphamide/dexamethasone immunosuppressed hamsters, and in STAT2-deficient transgenic hamsters. However, mortality was delayed compared to the parental ANDV strain. In STAT2-deficient animals, genomes from lung-isolated virus were not distinct from input attenuated strains, suggesting virus did not revert to the non-attenuated ANDV viral genome. Together these data indicate that the E265K mutation alters the ability of the virus to evade a normal immune response. The attenuated viruses appear to trigger protective innate responses because inoculation with mutant viruses three days before challenge protects hamsters against lethal disease caused by wild-type ANDV. Understanding how changes in the amino acid at position 265 can affect the innate response, and subsequent virulence of the infection, is an area of active research.

Keywords; attenuation, immunosuppressed, pathogenesis, innate immunity

● A novel three-dimensional vessel-on-chip model to study orthohantavirus pathogenesis

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Upon respiratory transmission via aerosolized fomites, Puumala orthohantavirus (PUUV) infection in humans can result in nephropathia epidemica. Endothelial cells (ECs) are predominantly infected, with EC activation and increased vascular permeability likely to play a central role in pathogenesis. Historically, most studies included ECs cultured in static two-dimensional (2D) conditions, which are physiologically not representative due to their lack of shear stress and their inherent pro-inflammatory state. Here, we present a novel three-dimensional (3D) primary umbilical vein endothelial vessel-on-chip microfluidic model for studying orthohantavirus infection. We established a high-throughput method for culturing vessels-on-chip in which we compared transcriptomic profiles of these ECs compared to static 2D-cultured ECs. RNAseq analyses demonstrate increased activity related to cytokines, growth factors and lipids by ECs cultured in vessels-on-chip. Upon TNF α stimulation, EC activation in these vessels leads to increases of inflammation marker ICAM-1 expression, monocyte adhesion and vascular permeability. PUUV infection causes upregulation of cxcl10, ido1 and lgals3bp genes expression, of which encoded proteins are known markers for disease severity. Additionally, PUUV infection results in increased monocyte adhesion without directly causing vascular permeability increases. This study characterizes a novel 3D vessel-on-chip platform to study orthohantavirus pathogenesis and highlights the intrinsic benefits of utilizing this novel model compared to conventional culturing methods. Additionally, this platform can be expanded by addition of co-cultures with other (immune) cell types to optimize representation of organ-specific microenvironments. Ultimately, this platform will pave the way for studying pathogenesis or assessment of possible therapeutics for a wide range of endotheliotropic viruses.

Key Words; Puumala orthohantavirus, vessel-on-chip, endothelium, pathogenesis

● A human lung xenograft mouse model of pan-orthohantavirus infection

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Orthohantaviruses primarily transmit and infect humans via the respiratory tract. While endothelial cells are an important target during later stages of infection, the initial target cells in the respiratory tract remain unknown. To date, studies on the molecular mechanisms of orthohantavirus-mediated pathogenesis have been hampered by the lack of biologically relevant models and small animal models susceptible for different orthohantavirus species. Thus, there is an urgent need for these models to study infection of the human respiratory tract. Here, we describe a novel human lung xenograft model in mice to study the pathogenesis and host responses of different orthohantavirus species infections within the same in vivo model, as well as pre-clinical applications of orthohantavirus therapeutics. Following implantation, human fetal lung xenografts rapidly develop mature structures of adult lungs including cartilage, blood vessels, ciliated pseudostratified columnar epithelium, and primitive "air" spaces filled with mucus and lined by cuboidal to flat epithelium. Following infection, distinct orthohantavirus species differentially replicate in the human lung and also disseminate to murine tissues. Orthohantaviruses primarily target the endothelium although other cell types, like macrophages and respiratory epithelium, are also infected. Infection occurs in the absence of overt histopathological changes. In addition, orthohantavirus dissemination can be blocked by treatment with a neutralizing antibody. This model will facilitate progress in the fundamental understanding of orthohantavirus pathogenesis and virus-host interactions. Moreover, this is a valuable model for pre-clinical evaluation of pan-orthohantavirus intervention strategies and provides a biologically relevant model for other viruses targeting the human respiratory tract.

Key Words; mouse model, pan-orthohantavirus animal model, human lung, monoclonal antibody, intervention

● A pro-inflammatory IgA response in acute Puumala Orthohantavirus-caused Hemorrhagic Fever with Renal Syndrome

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Immunoglobulin A (IgA) is the most abundant antibody isotype in humans. It is primarily found in bodily secretions and mucosal surfaces and thus strongly influences the course of respiratory tract viral infections. Despite its essential role, the contribution of IgA in acute viral diseases remain understudied, with the exception of recent reports suggesting it to contribute to COVID-19 pathogenesis. Human IgA comes in two forms, IgA1 and IgA2, which can exist as monomers, dimers or multimers. The Fc part of secreted IgA (sIgA) carries secretory component (SC).

In the current study, we aimed at describing the global IgA responses in two diseases caused by viruses entering through the respiratory tract: Hemorrhagic Fever with Renal Syndrome caused by Puumala Orthohantavirus and COVID-19 caused by SARS-CoV-2. In both diseases, we found elevated sIgA levels in acute disease patient blood samples, which returned to baseline levels in recovery. The sIgA levels associated with virus neutralization capability, polymericity and polyreactivity of IgA, all of which are attributes that waned rapidly at recovery. Importantly, virus-induced polymeric IgA showed significantly increased binding affinity to IgA Fc receptor, present primarily on myeloid cells, and activated pro-inflammatory responses in neutrophils in vitro. The latter findings suggest that the presence of "mucosal-type" polymeric IgA in blood of patients with acute viral infections is pro-inflammatory and may contribute to disease progression.

Key Words; IgA responses, mucosal immunity, IgA receptor, neutrophils, secretory component, COVID-19, Puumala Orthohantavirus, Hemorrhagic Fever with Renal Syndrome

● Two point mutations in protocadherin-1 afford protection against a lethal infection by Andes hantavirus

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New World hantaviruses, including Andes virus (ANDV) and Sin Nombre virus (SNV), are rodent-borne pathogens and the major etiologic agents of hantavirus cardiopulmonary syndrome (HCPS) in South and North America, respectively. Infection with these pathogens results in lethal human disease with fatality rates up to 40%. Limited knowledge of the viral life cycle, especially the mechanism of entry, has contributed to the absence of FDA-approved hantavirus-specific therapeutics or vaccines. In seminal work from this group, we identified a cadherin-superfamily protein, protocadherin-1 (PCDH1), as an entry receptor for HCPS-causing ANDV and SNV. Genetic depletion substantially reduced infection in vitro and PCDH1-knockout Syrian hamsters, generated by CRISPR/Cas9 genome engineering, survived a lethal ANDV challenge, highlighting the in vivo relevance of PCDH1 for pathogenesis and the therapeutic potential of this protein. Here, utilizing structure-based prediction, comprehensive site-directed mutagenesis, and protein-engineering, we mapped Gn/Gc's binding site onto PCDH1. Two amino acid sidechains within this site proved to be vital for PCDH1-mediated infection in vitro. The importance of these residues was validated in vivo, in which CRISPR knock-in mutant hamsters lacking these residues were significantly protected from a lethal ANDV challenge, solidifying their importance in vivo. Our findings point to a role for New World hantavirus Gn/Gc:PCDH1 recognition in influencing viral host range and also further define the critical molecular determinants within the entry host factor, PCDH1, that can be exploited to develop novel interventions to prevent and treat HCPS disease.

Key Words; hantavirus, protocadherin 1, host factors, Gn/Gc

Session4. Vaccines, Therapeutics & Clinical aspects

● High-Throughput Drug Repurposing Screen for Antivirals Targeting Orthohantaviruses

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Hantavirus disease causes tens of thousands annual cases with often high case fatality rates while no efficient vaccine or specific treatments are available.

In this project, we aim to identify antiviral compounds active against orthohantaviruses through a high-throughput screen of the SPECS Drug Repurposing Hub library: Over 5000 unique compounds that have previously reached clinical development were screened for substances that reduce the infectivity and/or replication of Puumala virus (PUUV) in A549 cells. The infection rate and level of viral proteins were determined via automated immunofluorescence microscopy.

We identified 152 compounds that decreased the infection rate with PUUV by at least 50% and 23 compounds that lead to an increase in infectivity of more than 100% without affecting cell viability. Several compounds affected the same pathways with mTOR/PI3K and HSP90 being prominent targets for viral inhibitors. Inhibition of different histone deacetylases commonly increased the infection rate. Other promising hits include integrin antagonists, protease inhibitors and compounds that regulate the cytoskeleton.

After validating our preliminary results, the effects of the hit compounds will be tested on other pathogenic hantaviruses in a BSL-3 facility. In parallel, additional experiments will be performed to further study the mechanism of how the compounds affect viral infection and replication and which role their targets play in the hantavirus life cycle.

Keywords: antivirals, drug repurposing, high-throughput drug screen, hantavirus

● Neutralizing Antibody Titers in Hospitalized Patients with Acute Puumala Orthohantavirus Infection Do Not Associate with Disease Severity

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Nephropathia epidemica (NE), a mild form of haemorrhagic fever with renal syndrome (HFRS), is an acute febrile illness caused by Puumala orthohantavirus (PUUV). NE manifests typically with acute kidney injury (AKI), with a case fatality rate of about 0.1%. The treatment and management of hantavirus infections are mainly supportive, although neutralizing monoclonal antibodies and immune sera therapeutics are under investigation. In order to assess the potential use of antibody therapeutics in NE, we sought to determine the relationship between circulating PUUV neutralizing antibodies, PUUV nucleocapsid protein (N) IgG antibodies, and viral loads with markers of disease severity. The study included serum samples of extensively characterized patient cohorts (n = 116) from Tampere University Hospital, Finland. The results showed that upon hospitalization, most patients already had considerable neutralizing and anti-PUUV-N IgG antibody levels. However, contrary to expectations, neutralizing antibody titers from the first day of hospitalization did not appear to protect from AKI or correlate with more favorable disease outcomes. This indicates that further studies are needed to investigate the applicability of neutralizing antibodies as a therapy for hospitalized NE patients

Key Words: nephropathia epidemica, AKI, pFRNT, disease severity, neutralizing antibody

● A Novel Disease Model for Sin Nombre Virus Infection in STAT2 Knockout Syrian Golden Hamsters

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Sin Nombre Virus (SNV) and Andes virus (ANDV) are New World hantaviruses (NWH) that are endemic in North and South America, respectively. They are the etiological agents of hantavirus cardiopulmonary syndrome (HCPS), which has a 40% case-fatality rate. SNV is a threat to public health and national security because of its broad reservoir pool and airborne transmission. There is no FDA-approved therapeutic or vaccine against SNV and rodent models are limited. The Syrian golden hamster is the only rodent species that develops HCPS-like symptoms and recapitulates human disease. This is specific to ANDV and Maporal virus (not known to cause human disease). Alternatively, SNV-infected hamsters quickly clear infection and are asymptomatic. Brocato et al., described lethal SNV infection in hamsters treated with dexamethasone and cyclophosphamide, which inhibit innate and adaptive immunity, respectively. Caveats of this model include a burdensome drug regimen and inhibition of the adaptive immune response, limiting its ability to be used in vaccine studies. Our group recently demonstrated that SNV causes symptomatic infection and 40% lethality in STAT2 knockout hamsters. These animals are unable to mount an interferon-mediated antiviral response and when infected with SNV they display viremia, respiratory distress, weight loss, and lung pathology. The current work characterizes, for the first time, the importance of the interferon response in lethal and symptomatic SNV infection in hamsters and provides a novel platform to test SNV-specific vaccines and therapeutics in vivo.

Key words: HCPS, SNV, hamsters, vaccines, therapeutics

● Validation of an antigenic site targeted by monoclonal antibodies against Puumala virus

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Identification of B-cell epitopes facilitates the development of vaccines, therapeutic antibodies and diagnostic tools. Earlier the binding site for bank vole monoclonal antibody (mAb) 4G2 against Puumala virus (PUUV, genus *Orthohantavirus*, family *Hantaviridae*, order *Bunyavirales*) was predicted by peptide display, phage-display and site-directed mutagenesis of vesicular stomatitis virus (VSV) particles pseudotyped with Gn and Gc glycoproteins of PUUV, which identified escape mutation F915A. To our surprise, a recent crystal structure of the Fab 4G2-PUUV Gc complex revealed that residue F915 is distal from epitope of mAb 4G2. To clarify this issue and find possible reason for the discrepancy, we designed a mutagenesis experiment to probe the structure-based epitope, and produced three mutant VSV pseudotypes E725A, S944F, S946F. These amino acid changes were able to convey neutralization escape to VSV pseudotypes, and confirm the functional relevance of this region for bank vole mAb 4G2 binding. Furthermore, amino acid changes S944F and S946F also convey escape from neutralization by human mAb 1C9. Identification of a common B-cell epitope for antibodies from different mammalian species might be of interest for subunit vaccine development against PUUV in future.

Keywords: monoclonal antibody, neutralizing epitope, site-directed mutagenesis.

● Persistence and Long-term Efficacy of Neutralizing Antibody Treatment Against Puumala Orthohantavirus in the Bank Vole Infection Model

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The Puumala orthohantavirus (PUUV) is a zoonotic pathogen carried by bank voles (*Myodes glareolus*) causing thousands of human cases of hemorrhagic fever with renal syndrome (HFRS) annually in Europe with no PUUV-specific treatment available to treat patients. Here we describe the long-term impact of treatment of PUUV-infected bank voles with a previously identified human neutralizing antibody (nAb). We show significant reductions in viral RNA amounts in treated animals still 14 days after infection despite many animals showing no traces of nAb in their serum at that time point. Time of nAb administration was found to have an impact at an early time point, but this impact waned over time. Most animals were not able to clear the virus even when receiving a second dose to boost serum nAb levels, which is in line with observations from naturally infected bank voles harboring PUUV despite high levels of neutralizing antibodies. Our results further characterize the bank vole as an animal model for PUUV in the context of antibody treatment, which will benefit future studies seeking to develop therapeutics against PUUV.

Key Words; Puumala virus, orthohantavirus, bank vole, monoclonal antibody, animal model

● Clinical and laboratory parameters of hantavirus patients in the Czech Republic

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Of the hantaviruses pathogenic for humans, Puumala (PUUV) and Dobrava (DOBV) viruses, namely its genotypes Dobrava (DOBV-D) and Kurkino (DOBV-K), are endemic in the Czech Republic. As causative agents of human infections, they are rarely demonstrated, but cause severe diseases even with rare deaths.

In the period 2008-2018, a total of 1761 patient samples from 3 regions were examined for hantaviruses in the Department of Virology, of which 61 persons were found to have hantavirus infection by serological methods, and 34 were also confirmed by PCR and sequencing.

The patients were predominantly male, with 50 males and 11 females aged 13-71 years. DOBV was detected in 55 patients, PUUV in 4 patients, Tula virus in 1 patient and in one patient the virus was not identified.

Of the 34 PCR positive samples, DOBV was confirmed 31 times (17 times DOBV-D, 14 times DOBV-K), 2 times PUUV and 1 times Tula.

We had clinical and laboratory data from 30 patients with hantavirus infection confirmed by PCR and sequencing. Of these, 27 were DOBV (16 DOBV-D and 11 DOBV-K), 2 PUUV and 1 Tula. We statistically processed the data from patients with DOBV infection and compared them for both genotypes (DOBV-D and DOBV-K).

Significant differences ($p < 0.05$) were found in the following parameters: blood platelet values were higher in DOBV-K, while blood leukocytes, serum aspartate aminotransferase (AST), procalcitonin (PCT) and urea levels were higher in DOBV-D patients. There were no significant differences in serum C-reactive protein (CRP), alanine aminotransferase (ALT) and creatinine levels between the two genotypes. The length of hospitalization as well as the number of days spent in ICU were significantly higher in patients with DOBV-D infection.

The following symptoms occurred with significantly higher frequency in DOBV-D patients than in DOBV-K patients: nausea, vomiting, diarrhoea, bleeding, macroscopic haematuria, pancreatic irritation and need for haemodialysis. For the other symptoms compared, there were no significant differences between the two genotypes (headache, back and abdominal pain, renal insufficiency, proteinuria, microscopic haematuria, respiratory symptoms). Some symptoms occurred only in DOBV-D patients, while none occurred in DOBV-K patients, but statistical significance was not reached due to the small number of individuals (acute myopia, conjunctivitis, exanthema and artificial pulmonary ventilation).

Overall, the course of the disease was judged to be more severe in patients with DOBV-D, whereas the course was less severe in patients with DOBV-K infection. Death occurred in only 1 patient with DOBV-D.

Session5. Diagnosis & Emerging Hantaviruses

● The Identification of A Novel Etiological Agent of Hemorrhagic Fever with Renal Syndrome in Southern Areas, The Republic of Korea Using Multiplex PCR-based MinION Sequencing

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Rodent-borne hantaviruses, single-stranded negative RNA viruses, cause severe and fatal human infectious diseases including hemorrhagic fever with renal syndrome (HFRS) and hantavirus cardiopulmonary syndrome. Defining the genome sequences of hantaviruses plays a critical role in the development of preventative and therapeutic measures. In a previous study, a new genotype of the Hantaan orthohantavirus (HTNV) was discovered in *Apodemus agrarius chejuensis* (*A. chejuensis*) on the Jeju Island, Republic of Korea (ROK). However, whether this genotype could cause HFRS in humans remains to be investigated. To examine this potential, a total of 196 samples were collected from HFRS patients and *A. agrarius* in the southern mainland of ROK. In nature, the positivity of southern HTNV genotypes was shown up to 32% by performing reverse transcription-polymerase chain reaction (RT-PCR). Multiplex PCR-based MinION sequencing elicited nearly whole-genome sequences of novel HTNV genotypes from clinical and natural host samples. The phylogenetic pattern of HTNV genome sequences, recovered from five *A. agrarius* and six HFRS patients, shaped the genetic lineage with HTNV strains found in *A. chejuensis* on the Jeju Island. These findings suggest that the novel genotype of HTNV has the potential to infect humans and induce HFRS. As a result, the study provides the understanding and need for preemptive precautions and awareness of newly emerging HTNV genotypes among physicians in the southern region of ROK.

Key Words; novel genotype of HTNV, HFRS, southern areas of ROK, multiplex PCR-based MinION sequencing

● Identification and Complete Coding Sequence of Limestone Canyon Virus in Wild-Caught *Peromyscus boylii* from New Mexico, USA

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Orthohantaviruses are carried and transmitted by different small mammal reservoirs throughout the world. In the United States (US), the primary rodent reservoir for pathogenic hantaviruses is the western deer mouse, *Peromyscus maniculatus sonoriensis*, that carries Sin Nombre virus (SNV). New Mexico is the leading state for SNV infection that causes Hantavirus Cardiopulmonary Syndrome (HCPS) in the US. However, multiple species of *Peromyscus* have been reported to carry different orthohantaviruses, such as New York virus (*Peromyscus leucopus*) and Limestone Canyon virus (*Peromyscus boylii*). Although multiple *Peromyscus* species are known to reside in New Mexico, only SNV has been found.

Using a set of pan-orthohantavirus primers, we detected an orthohantavirus other than SNV in a *P. boylii* captured in 2019. We used whole-genome sequencing to generate a consensus genome. Phylogenetic analysis of the genome was performed and showed close similarity to published partial sequences of Limestone Canyon virus (LSCV). A set of primers spanning each segment was developed for sequencing and confirmation. A previous unpublished study of wild-caught rodents from a different region of New Mexico in 2006 found LSCV evidence in a population of *P. boylii*. The combined data revealed similarity between our complete genome and partial fragments of previously reported LSCV while identifying two clades based on geographic location and host subspecies. Here we report for the first time a full reference genome for LSCV and the discovery of its presence in multiple regions in New Mexico.

Keywords; New Mexico, *Peromyscus boylii*, host reservoirs, sequencing, Limestone Canyon virus

● Urinary Genome Detection and Surveillance of Hantaan Orthohantavirus from Hemorrhagic Fever with Renal Syndrome Patients Using Amplicon-based Next-generation Sequencing

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Hantavirus infection occurs through the inhalation of aerosolized particles, including urine, feces, and saliva of infected rodents. The presence of Hantaan virus (HTNV) RNA or infectious virions in urine specimens from a patient with hemorrhagic fever with renal syndrome (HFRS) has remains unclear. We performed multiplex PCR-based next-generation sequencing (NGS) to acquire the genomic sequences of HTNV in urine and serum specimens from four Republic of Korea Army patients with HFRS. The epidemiological and phylogenetic analyses showed geographically homogenous lineage with those in *Apodemus agrarius* captured in highly endemic areas. These findings suggest that phylogeographic tracing of HTNV genomes reveals the potential infection exposure sites of patients with HFRS. Genome reassortment analyses demonstrated a genetic configuration compatible with HTNV L segment exchange in nature. These results suggest that genome-based diagnosis of HTNV from the urine enabled to track the putative infection sites of patients with HFRS by phylogeographically linking to the zoonotic HTNV from the reservoir host captured at endemic regions. Our findings raise awareness among physicians for the presence of HTNV in the urine of patients with HFRS.

Keywords; Hantaan orthohantavirus (HTNV), Urinary genome detection, Hemorrhagic fever with renal syndrome(HFRS), Multiplex PCR, Next-generation sequencing

● Use of a New World hantavirus RT-qPCR assay to detect multiple hantavirus in the Americas

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Hantavirus pulmonary syndrome (HPS) cases and non-HPS hantavirus infections are typically overlooked in the Americas. Multiple New World hantaviruses have been reported, however, hantaviruses remain underdiagnosed due to lack of awareness and testing in some endemic areas. Serological assays have been previously used to diagnose hantavirus. As part of COVID-19 pandemic response, the capacity to perform molecular assays has significantly increased in the Americas. The Pan American Health Organization and collaborative centers have identified a necessity to validate hantavirus molecular assays that could be implemented in the region. With that objective, a New World hantavirus (N gene) RT-qPCR TaqMan assay developed at CDC, was tested using multiple New World hantavirus strains in our collection including Andes, Black Creek Canal, Convict Creek, Lechiguanas, Laguna Negra, Maciel and Sin Nombre virus, as well as the Old World hantavirus, Seoul, Hantaan and Puumala viruses. The RT-qPCR detected all New World hantaviruses and did not detect Old World hantaviruses. In addition, hantavirus RNA was identified using this RT-qPCR in human samples collected in Bolivia in 2018 and 2019. Of 14 patients with available clinical data, IgM and IgG antibodies were identified in 12 (86%) and 7 (15%) blood specimens; 11 were RT-qPCR positive. Sequencing provided preliminary identification of hantaviruses circulating in Bolivia as Oran , Tunari-like, Alto Paraguay-like, Laguna Negra, and Lechiguanas-like from various geographic locations. We report here the New World hantavirus RT-qPCR results and the phylogenetic tree from Bolivia. This work will help improve monitoring of hantavirus circulating in the Americas.

Poster Presentations

Session1. Viral epidemiology, evolution & genetics

● Evidence of orthohantavirus and leptospira infections in small mammals in an endemic area of Gampaha district in Sri Lanka

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The epidemiology of orthohantavirus infections and *leptospirosis* is similar and presents related clinical pictures in humans. However, a paucity of data on actual reservoir hosts for orthohantaviruses and leptospira exists. Therefore, this study aimed at determining the occurrence of orthohantaviruses and leptospira in small mammals captured in an endemic region of Sri Lanka.

Animals were identified using morphological keys and DNA barcoding techniques targeting the cytochrome oxidase b subunit gene (Cytb). Lung tissues and sera were subsequently analyzed for the presence of orthohantavirus RNA using qRT-PCR and antibodies against these pathogens.

Forty-three (43) animals representing: *Rattus*; *R. rattus* (black rat) or *R. tanezumii* (Asian rat), *R. norvegicus* (brown rat), *Suncus murinus* (Asian house shrew), and *Mus musculus* (house mouse) were investigated. No orthohantavirus RNA was detected from the lung tissue or serum samples of these animals. IgG antibodies against Puumala orthohantavirus (PUUV) and/or Seoul orthohantavirus (SEOV) antigens were detected in sera of 28 (72%) out of the 39 rats analyzed. Interestingly, sera of 36 (92%) out of 39 rats also showed presence of anti leptospira-IgG antibodies, representing dual infection or dual exposure in 26/39 (66.7%) of examined rats.

Most rats (72%) in our study displayed antibodies reacting to orthohantavirus NP antigens, related to PUUV and/or SEOV. No correlation between the orthohantavirus and leptospira IgG antibody levels were noticed. Finally, a combination of both morphological and DNA barcoding approaches revealed that several species of rats may play a role in the maintenance and transmission of orthohantavirus and leptospira in Sri Lanka.

Keywords; Sri Lanka, orthohantavirus, leptospira, rodents, shrews

● Hantavirus infection as a risk factor for chronic kidney disease of unknown etiology (CKDu) in Sri Lanka

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Chronic kidney disease of unknown aetiology (CKDu) is a major public health problem in Sri Lanka, particularly in agrarian communities. Although the cause of CKDu remains unknown, we propose hantavirus infection as a risk factor. Based on seroepidemiological analysis between CKDu patients and healthy residents in endemic and non-endemic areas, a significant relationship was found between CKDu and hantavirus seropositivity. In addition, two species of hantavirus: Lanka virus carried by *Mus booduga*, and Anjozorobe virus carried by *Rattus rattus* were found in CKDu endemic areas. However, it is unknown which viruses infect humans. In addition, CKDu in Sri Lanka has emerged since the early 1990s. It is not known when the high rate of human infection began in Sri Lanka.

In this study, we analyzed sera collected from CKDu-endemic areas, Girandurukotte, and Medawachchiya by 2010. A high seroprevalence among CKDu patients has already been observed in these sera. This result indicated that approximately half of the CKDu patients had a history of hantavirus infection in 2010. We further established a novel diagnostic method based on two novel hantavirus sequences detected in rodents. We believe that the Lanka virus is responsible for human hantavirus infection in Sri Lanka. It is important to clarify how hantavirus infection is associated with CKDu progression.

Key Words; Thailand virus, Lanka virus, Indian field mouse, Sri Lanka

● Widespread presence of Brno hantavirus in common noctule bats (*Nyctalus noctula*) in Central Europe

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Initially, hantaviruses were thought to be exclusively rodent-borne. Nowadays, several shrew-, mole- and bat-borne hantaviruses were described. Until the discovery of Brno hantavirus (BRNV) in common noctule bats in Brno, the Czech Republic, Europe, the bat-borne hantaviruses were detected and characterized exclusively from Africa and Asia. In this study we aimed to evaluate the presence, prevalence and host specificity of BRNV through a field survey of bats from Central Europe.

Bats and/or bat swab samples from Poland, Germany and Austria were subjected to RNA isolation and conventional RT-PCR and a novel BRNV-specific real-time RT-qPCR both targeting large (L) segment of BRNV. Screening of organ pool samples (1047 bats of 21 species) from Germany resulted in three positive samples. From Austria, out of 618 swabs and 155 pooled tissue samples collected from 464 bats of 18 species, one tissue sample was positive. Of the 77 bats belonging to 12 species from Poland, three samples were positive. All positive samples originated from common noctule bats (*Nyctalus noctula*). Phylogenetic analysis confirmed high similarity between sequences from positive samples from Germany and Poland and two sequences from the Czech Republic.

This study detected for the first time BRNV in common noctules from Germany, Poland and Austria. The multiple detection of BRNV in the same bat species from different regions, together with the absence of BRNV-specific RNA in other sympatric bat species, indicated its clear host specificity. The study was supported by the grant project no. NU21-05-00143 from the Ministry of Health of the Czech Republic.

Key Words; hantavirus, bats, RT-qPCR, Europe, host specificity

● Phylogeny of Shrew- and Mole-Borne Hantaviruses in Poland and Ukraine

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Earlier, we demonstrated the co-circulation of genetically distinct non-rodent-borne hantaviruses in central Poland, including Boginia virus (BOGV) in the Eurasian water shrew (*Neomys fodiens*), Seewis virus (SWSV) in the Eurasian common shrew (*Sorex araneus*) and Nova virus (NVAV) in the European mole (*Talpa europaea*). To further investigate the phylogeny of hantaviruses harbored by soricid and talpid reservoir hosts, we analyzed RNAlater®-preserved lung tissues from 320 shrews and 26 moles, captured during 1990–2017 across Poland, and 10 European moles from Ukraine for hantavirus RNA by RT-PCR and DNA sequencing. SWSV and Altai virus (ALTV) were detected in *Sorex araneus* and *Sorex minutus* in Boginia and Białowieża Forest, respectively, and NVAV was detected in *Talpa europaea* in Huta Dłutowska, Poland, and in Lviv, Ukraine. Phylogenetic analyses, using maximum-likelihood and Bayesian methods, showed geographic-specific lineages of SWSV in Poland and elsewhere in Eurasia and NVAV in Poland and Ukraine. The ATLTV strain in *Sorex minutus* from Białowieża Forest on the Polish-Belarusian border was distantly related to the ATLTV strain previously reported in *Sorex minutus* from Chmiel in southeastern Poland. Overall, the phylogeny supported long-standing host-specific adaptation. The recent discovery of Academ virus in the Siberian mole (*Talpa altaica*) in Russia and Asturias virus in the Iberian mole (*Talpa occidentalis*) in Spain would predict that other genetically distinct hantaviruses are harbored by mole species belonging to the genus *Talpa*. Finally, studies are warranted to map the geographic distribution and genetic diversity of non-rodent-borne hantaviruses in Ukraine.

Key Words; *Hantaviridae*; hantavirus; shrew; mole; Poland; Ukraine

● Brno loanvirus (BRNV) in bats inhabiting urban area of Brno, Czech Republic

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Pathogenic hantaviruses (family Hantaviridae) represent an emerging group of viruses with several members that are of medical importance. Bats are already known to harbor a variety of emerging pathogens, including hantaviruses. Their ability to fly and their social lifestyle allow for efficient maintenance, evolution, and spread of pathogens. A novel hantavirus, termed Brno loanvirus (BRNV; subfamily Mammantavirinae), was recently detected in the Central European bat species *Nyctalus noctula* and its complete sequences of all coding regions were determined in our previous study. In this study, we investigated the presence of BRNV in a group of bats originating from the urban area of the city of Brno.

A total of sixty-three bats representing four species were included in the screening study: *Nyctalus noctula* (n=58), *Pipistrellus pipistrellus* (n=2), *Myotis nattereri* (n=2), *Myotis brandtii* (n=1). The presence of BRNV was tested by nested RT-PCR using RNA isolated from liver homogenate. Degenerated oligonucleotides were used to detect a 412 bp long PCR product corresponding to the fragment of RNA-dependent RNA polymerase. Of the sixty-three samples tested, ten samples tested positive for the presence of the PCR product, which was subsequently sequenced to confirm the specificity of the assay. All bats that tested positive belonged to the species *N. noctula*. BRNV was quantified by RT-qPCR in various bat organs. We also intended to isolate BRNV from the tissues of the positive bats *N. noctula*. The results of this study confirm that *N. noctula* can serve as a reservoir host for Brno loanvirus even in urban areas.

The study was supported by the grant project no. NU21-05-00143 from the Ministry of Health of the Czech Republic.

Key Words; Brno loanvirus (BRNV), bats, RT-qPCR

● **A novel genotype of Hantaan virus from *Apodemus chejuensis* from Jeju Island as a potential etiologic agent of hemorrhagic fever with renal syndrome in Republic of Korea**

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Hantaan orthohantavirus (HTNV) is a major causative agent of hemorrhagic fever with renal syndrome (HFRS). According to Korea Disease Control and Prevention Agency, about 18 HFRS cases have been reported over the past decade on Jeju Island, Republic of Korea (ROK). However, pathogenic hantaviruses to humans have not been detected from natural reservoir hosts collected on Jeju Island. We performed the targeted rodent trapping on Jeju Island, ROK, during 2018-2020. Using indirect immunofluorescence assay test and reverse transcription-polymerase chain reaction, the anti-HTNV IgG antibody and RNA were detected from autochthonous Jeju striped field mouse (*Apodemus chejuensis*). Amplicon-based next generation sequencing (NGS) was conducted to obtain whole-genome sequences of HTNV. The phylogeny and tanglegram interference were generated to compare different phylogeographic links matching between hantaviruses and their reservoir hosts. The positivity of serological test was 14.1% (9/64). A total of six seropositive mice harbored HTNV RNA. The six complete genomes of HTNV were recovered from the *A. chejuensis* using amplicon-based NGS. Phylogenetic analysis demonstrated a novel distinct HTNV genotype. The detected HTNV genomic sequences were phylogenetically related to a viral sequence derived from HFRS patient in southern ROK. Tanglegram analysis demonstrated the segregation of HTNV genotypes corresponding to *Apodemus* spp. divergence. Our results suggest that HTNV harbored by *A. chejuensis* may be a potential etiological agent of HFRS in southern ROK. Ancestral HTNV may infect *A. chejuensis* prior to geological isolation between the Korean peninsula and Jeju Island, supporting the co-evolution of orthohantaviruses and their reservoir hosts. These findings arise awareness among physicians for HFRS outbreaks in southern areas, ROK.

Key Words; Hantaan virus(HTNV), novel genotype, Hemorrhagic fever with renal syndrome(HFRS), *Apodemus chejuensis*, Jeju Island

● A Potential Relationship Between Sin Nombre Hantavirus and the Microbiome of the Deer Mouse, *Peromyscus maniculatus*

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Sin Nombre Hantavirus (SNV) is a causal agent of hantavirus disease in North America, first identified in the four corners region of the United States in 1993. SNV infection primarily occurs in the Southwestern U.S.A. SNV causes hantavirus cardiopulmonary syndrome which has a case fatality rate of approximately 35%. The primary reservoir of SNV is the deer mouse, *Peromyscus maniculatus*. Despite high and widespread SNV infection rates in deer mice (approximately 25%), human cases remain rare and geographically isolated. The reason for this isolation of human cases remains unknown.

Microbiome research in mosquitoes has indicated that colonization with certain symbiotic bacteria reduces the ability of mosquitos to transmit various arboviruses to new hosts, and that colonization with these bacteria reduces the efficiency of viral replication and transmission. Knowing this, we investigated the microbiome of deer mice captured in New Mexico in the Southwestern United States. A pilot study was performed using feces from 24 wild-caught animals from a single site that were stratified based on age, sex, and infection status as determined by qRT-PCR probing for the SNV S segment. It was found by sequencing the hypervariable region of bacterial 16s rRNA, that while the microbiomes studied did not group based on any mouse physiological measures, the SNV positive microbiomes were found to group together and were less diverse. This indicates there is an interaction between the microbiome and SNV and this potential relationship bears further study.

Key words; microbiome, Sin Nombre Virus, *Peromyscus maniculatus*, new-world hantavirus

● Two Divergent Hantaviruses, Biya River and Boginia, in the Eurasian Water Shrew (*Neomys fodiens*) in Siberia

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Introduction. Insectivores and moles are a new reservoir of hantaviruses. Several pairs of highly divergent hantavirus species have been identified among the same species of shrews and moles. Seewis (SWSV) and Altai (ALTV) viruses harbored by the Eurasian common shrews, Nova and Bruges viruses harbored by the European mole are examples of host sharing in Eurasia.

Methods. A total of 31 Eurasian water shrews (*Neomys fodiens*) were captured during 2019-2022 in 3 localities in Western Siberia. Lung tissues were analyzed for hantavirus RNA using RT-PCR. Viral RNA was detected in 4 out of 31 *Neomys fodiens*. Sanger and Illumina techniques were used for partial or complete genome sequencing of these wild type strains.

Results. Pairwise alignment and comparison of partial- and nearly complete genome sequences from *Neomys fodiens*, as well as phylogenetic analyses, using maximum likelihood method indicated that a distinct hantavirus species were identified. One of these hantaviruses was Boginia virus (BOGV), previously detected among *Neomys fodiens* in Poland and Finland. Here, we identified a new hantavirus, designated Biya River virus (BIRV), detected in the Eurasian water shrew, captured near Teletskoye Lake in the Altai Republic. The nearly complete genomic sequence analysis showed that BIRV had a common origin with ALTV and appears to belong to the Mobatvirus genus, while BOGV is a member of the Orthohantavirus genus. These findings suggest that BIRV might have emerged from ancient cross-species transmission.

Key words: hantavirus, shrew, evolution, host-switch, Russia

● Development of a Seroepidemiological Tool for Bat-Borne and Shrew-borne Hantaviruses and Its Application for Samples from Zambia

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Apart from the classical disease-causing rodent-borne hantaviruses caused by Hantaan and Sin Nombre viruses, numerous species of hantaviruses of shrews and bats have also been reported and their pathogenicity were not examined. Due to the difficulty in hantavirus detection, this study focused on developing a serological detection method for detecting antibodies to bat- and shrew-borne hantaviruses. Eleven viruses (Bat-borne: Brno, Dakrong, Quezon, Robina, Xuan Song and Shrew-borne: Asama, Altai, Cao Bang, Nova, Seewis and Thottapalayam) were selected based on the phylogenetic analysis of their N proteins difference. The recombinant N (rN) proteins of these viruses were expressed as antigens in the Vero E6 and 293T cell lines using the pCAGGS/MCS vector. Polyclonal antibodies to the NUS tag- rN fusion protein of these viruses were also generated in mice and rabbits as positive control antisera. Five expression plasmid vectors for rN of bat-borne viruses were transfected together to produce mixed rN antigen-expressing Vero E6 cells. Similarly, mixed rN of shrew-borne viruses was prepared. Following confirmation of all positive controls, bat and shrew mixed antigens were used for screening multiple samples for antibody detection in the indirect immunofluorescent antibody test (IFAT) using anti-bat IgG (Bethyl Laboratories) as a secondary antibody. Positive samples were then individually tested for the respective antigens in IFAT and immunoblotting assays.

Of 1764 bat sera samples from Zambia, 400 positive sera were found, with 9.1% and 17.1% for bat and shrew mixed antigen systems, respectively. Furthermore, individual IFAT and the immunoblotting assays showed positive reactions to Brno, Dakrong, Quezon, Xuan Son, Asama, Altai, and Thottapalayam viruses. These observations suggest that the mixed antigen screening system is useful for serological screening. In addition, bats in Zambia are potential hosts for hantaviruses.

Key Words; single expression system, bat and shrew-borne hantaviruses, mixed antigen system

● Estimating time-scale of hantavirus evolution

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Known to cause haemorrhagic fever with renal syndrome in Eurasia and hantavirus cardiopulmonary syndrome in the Americas, hantaviruses remain globally emerging pathogens. Genetic drift, homologous recombination and reassortment have been proposed as main mechanisms of hantaviral genetic diversity. Hantavirus evolution was previously considered inextricably linked with the evolution of their hosts, but this relationship has more recently been vigorously debated, whereas the timescale of viral diversification events remains elusive. In this study phylogenetic analyses were used to estimate the divergence time of the five main Eurasian hantaviral species: Puumala (PUUV), Seoul virus (SEOV), Hantaan virus (HTNV), Dobrava-Belgrade (DOBV) virus and Tula virus (TULV).

The study was undertaken on the total of 902 complete coding nucleotide sequences of the S segment N gene. The alignment of 1299 nt in length comprised 194 HTNV, 359 PUUV, 103 SEOV, 59 TULV, 51 DOBV and 138 sequences of 27 other hantaviruses. Phylogenetic content of the alignment was assessed through Maximum likelihood analysis using TREE-PUZZLE 5.2 and found to be sufficient. Temporal structure of the dataset was evaluated by a regression of root-to-tip genetic distances against date of sampling using the Path-O-Gen program, revealing a limited yet present temporal structure, with positive correlation coefficient of 0.103. The time-scale analysis was performed in BEAST 1.10.4. under GTR+I+G substitution model. Divergence time of the five main Eurasian hantaviral species (PUUV, SEOV, HTNV, DOBV, TULV) was estimated in the range of 10000 years ago, whereas for the most recent common ancestor of all hantaviruses, probably 20000 years ago.

Key Words: hantavirus, evolution, divergence time

● Exploring the Molecular Evolution and Genetic Diversity of Hantaan and Seoul Viruses: A Comparative Genomic Analysis

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Hantaan virus (HTNV) and Seoul virus (SEOV) are zoonotic viruses that cause hemorrhagic fever with renal syndrome epidemic in humans. Although clinical symptoms caused by HTNV and SEOV are very similar, HTNV and SEOV exhibit different evolutionary dynamics. In this study, we performed phylogenetic and bioinformatic analyses using the complete genome sequences of the large (L), medium (M), and small (S) gene segments of HTNV and SEOV from a holistic perspective. Our analyses revealed that the phylogenies of each genome segment of HTNV and SEOV generally formed major genetic clades based on their geographical distribution. Notably, SEOV showed higher evolutionary rates and diversity in the M and S segments than HTNV did. In the genetic reassortment analysis, SEOV appeared to be more restrictive in the reassortment between the L and M genes than other genetic reassortment pairs whereas HTNV exhibited more stringent reassortment permissiveness between the M and S genes. Both viruses share similar putative glycosylation patterns in the Gn/Gc glycoprotein, but the M gene of HTNV harbored more genetic clade-defining mutations than that of SEOV. Especially, five lysine residues (K77, K82, K231, K307, and K310) located at the tip of the Gn and structural shift around the 200-helix may contribute to the distinctive antigenic properties of SEOV, compared to those of HTNV, potentially impacting the effectiveness of neutralizing antibodies. Our results reveal key molecular differences in the evolutionary dynamics of HTNV and SEOV, which may potentially contribute to the design and development of preventive vaccines against both viruses.

Hantaan virus; molecular evolution; phylogeny; reassortment; Seoul virus

● Hantavirus Infection During a Wild Rodent Population Outbreak “Ratada” in Southern Brazil

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Rodent outbreaks are known as the rapid growth of rodent populations due to a large food supply associated with mast seed events. In Brazil, there are only few reports of this phenomenon called “ratada”, most of them from the south region, associated with bamboo blooming. In October 2022, a rodent fieldwork was conducted after a Hantavirus Pulmonary Syndrome outbreak during a “ratada” in Urubici, state of Santa Catarina - southern Brazil. A mast-seeding of Taquara bamboos (*Merostachys* sp.) concomitant to Araucaria fruiting season was observed in the study area located in the Brazilian Araucaria Atlantic Rainforest.

Animals were sampled over two nights with a trapping effort of 400 trap-nights in five different localities. A total of 158 rodent specimens were live-trapped with an overall trapping success of 39.5%. Antibodies IgG against ANDV were detected in 10 Sigmodontinae rodent specimens belonging to *Akodon montensis*, *A. paranaensis*, *A. serrensis*, *Delomys dorsalis*, *Oligoryzomys nigripes* and *Thaptomys nigrita* species. Viral RNA was detected in *A. serrensis*, *D. dorsalis* and *O. nigripes*.

The increase in the abundance of sympatric rodent species seems to favour hantavirus spillover, highlighted by the first description of *D. dorsalis* naturally infected. Also, the only record of *T. nigrita* and *A. serrensis* infection has been reported during a rodent outbreak in south Brazil. As there are limited data on hantavirus dynamics during rodent outbreaks, further investigations on the population ecology of reservoir rodents are needed to better understand the “ratada’s” phenomenon and thus provide information for predicting and controlling HPS outbreaks.

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Keywords: ratada, rodent outbreak, hantavirus reservoir, HPS, brazil

● Is *Carollia perspicillata* a Natural Host of a Novel Hantavirus in the Brazilian Atlantic Rainforest?

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A diversity of hantavirus species has recently been observed in a variety of non-rodent and insectivores hosts, such as fish, reptiles and bats, expanding our understanding about nature and the origins of the hantaviruses. Bat-borne hantaviruses have been identified in different species of Old-World bats in Asia, Europe, Africa and Australia. So far, none have been detected in the New World bats. Brazil hosts approximately 15% of the world's bat diversity, with 181 distinct species, 68 genera and 9 families.

Here in, we tested 98 tissue samples from 9 bat species captured in the Biological Reserve of Guaribas, in Paraíba State, Northeastern Brazil, in 2015. Part of the hantaviral L segment (400pb) was amplified from the lungs of 7 frugivorous *Carollia perspicillata* bats. The Seba's short-tailed bat (*C. perspicillata*) has a wide geographic distribution, ranging from Mexico through Central and South America. The sequences from *C. perspicillata* showed over 82% identity with HantaV-1/hantavirus strains found in bats of the *Carollia* genus in different South American countries.

Although considering the recognized bat-borne hantaviruses, the Brazilian sequences revealed the highest similarity (approximately 75%) with *Brno loanvirus* specie, identified in the common noctule bat (*Nyctalus noctula*) from Europe. Further analysis are yet to be done, but our results suggest that the sequences described here are representative of a novel hantavirus attempt named "Guaribas virus" and points the Seba's short-tailed bat *C. perspicillata* as the natural host for this virus in South America.

Keywords: bats, hantavirus, *Carollia perspicillata*, atlantic rainforest, brazil

● First molecular evidence of shrew-borne Seewis orthohantavirus in Croatia

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For a long time, after the initial isolation of Hantaan orthohantavirus, the scientific consensus was that rodents were the only reservoirs for orthohantaviruses. It took several decades to detect and isolate the first non-rodent-borne orthohantaviruses from shrews, moles, and bats. In 2007 partly genome sequences of a new shrew-borne hantavirus, named Seewis virus, were amplified from the lung tissue of a single *Sorex araneus* (Eurasian common shrew) caught in Switzerland.

Thereafter, Seewis virus has been detected in several European countries, including Hungary and Slovenia, which border Croatia. As Croatia is a well-known endemic region for several zoonotic agents, including various hantaviruses, e.g., Dobrava orthohantavirus, Puumala orthohantavirus, and Tula orthohantavirus, it is safe to assume that Seewis virus is also endemic in Croatia.

Shrew samples trapped by accidents from our wild rodent collection from 2012 onwards have been tested by a Pan-Hantavirus RT-PCR for the presence of large RNA genome segment. Additional nested PCR systems were employed to amplify fragments of the other genome segment for phylogenetic analysis. In addition, the cytochrome B sequence was used to molecularly determine the host species.

Here we present the first molecular evidence and phylogenetic analysis of Seewis virus in Croatia. Of the tested shrews, two *Sorex araneus* were positive for L segment RNA. Further phylogenetic analysis of a partly small segment sequence showed an identity between 90,4% and 95% when compared to sequences from Slovenia, whereas more distant virus sequences had less than 90% sequence identity.

Key Words: seewis orthohantavirus, phylogeny, shrew-borne, *Sorey Araneus*

Session2. Replication, Transcription, Translation & Assembly

● Molecular Basis for the Host Adaptation of Zoonotic Orthohantaviruses

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Human-pathogenic orthohantaviruses are rodent-borne pathogens that do not cause obvious pathogenic effects within their natural hosts. Upon transmission to humans, European orthohantaviruses can cause hemorrhagic fever with renal syndrome of high severity. Despite the disease burden, our knowledge about the infection mechanism and the influence of adaptive amino acid exchanges within the envelope proteins remains incomplete.

Focusing on virus entry, our goal is to investigate orthohantavirus infection in human and rodent-derived cell models. We use a combination of both single-cell microscopy as well as proteomics to identify receptor candidates and to clarify the signaling pathways that are triggered by viral attachment and entry within the different cell lines. In parallel, we have started to comparatively characterize the different host cell lines with respect to their morphology and physiology, most importantly their endocytic pathways, by using selected orthohantavirus strain infections. The overall infectivity of viral strains will be quantified by different microscopy techniques. To increase the validity of our observations, we will complement our imaging results with quantification by flow cytometry. To correlate cell-type dependent effects with viral adaptation, a further focus of our work is to investigate the spatial and temporal evolution of the orthohantavirus glycoprotein complex (GPC), a main determinant of viral tropism. This study will build the basis to further elaborate the functional consequences of adaptative amino acid exchanges with the help of GPC-pseudotyped Vesicular Stomatitis Viruses (pVSV).

Key Words; cell susceptibility, cell response, virus-cell entry, host tropism, microscopy

● Analysis of relationship between replication of Hokkaido orthohantavirus and autophagy

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Hokkaido orthohantavirus (HOKV) is a member of viruses among the family Hantaviridae, and the virus genome consists of three-segmented RNA viruses with S, M, and L. However, the mechanism of virus replication is still unknown. HOKV is hosted by gray-red backed vole, *Myodes rufocanus*, and is genetically closely related to Puumala orthohantavirus (PUUV). Autophagy is a mechanism common to eukaryotic cells that degrades intracellular substances under stressful conditions. Autophagy is not only involved in host cell homeostasis, but is also known to be related to the replication of some viruses. In this study, we aimed to clarify the relationship between HOKV replication and autophagy.

[Materials and Methods]

MRK101 cells derived from the kidneys of *M. rufocanus* were infected with HOKV and the localization of nucleocapsid protein (NP) and LC3, an autophagosome marker, was observed by fluorescent antibody assay. The expression of Atg5, an autophagy related-gene, was suppressed with siRNA and then the cells were infected with HOKV. The viral RNA in the cells was quantified by RT-qPCR. HOKV-infected MRK101 cells were treated with GF-109203X, a known autophagy inducer, and the amount of viral RNA in the cells was quantified by RT-qPCR.

[Results]

Co-localization of NP and LC3 was observed in HOKV-infected MRK101 cells. The replication of HOKV increased in Atg5-suppressed MRK101 cells than in the control cells. On the other hand, the amount of viral RNA in the HOKV-infected cells significantly decreased by GF-109203X treatment.

[Discussion]

Since NP and LC3 were co-localized, it implies that HOKV NP localizes to autophagosomes. The amount of viral RNA increased in Atg5-suppressed cells while the amount of viral RNA in GF-109203X-treated cells decreased, suggesting autophagy suppresses HOKV replication.

Session3. Immune responses & Pathogenesis

● Epithelial Cell Proliferation in Kidneys and Relationship With Changes in Acid and Alkaline Phosphatases in Hemorrhagic Fever With Renal Syndrome

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Comparative analysis of renal morphology findings of 58 HFRS patients at different stages of the disease, using immunohistochemical and morphometric techniques, as well as proliferation of tubular epithelium and distribution of nonspecific phosphatases in apical and basal regions of nephrothelium and capillary endothelium was made.

At an acute stage within 10 days after the onset of the disease (20 cases), necrotic changes, granular and diffuse hydropic dystrophy in tubular epithelium were noted. Edema, focal hemorrhages and focal lymphoid-macrophage infiltrates in stroma were seen. Within up to 20 days after the onset of the disease (21 cases), thickening of basement membranes, focal segmental sclerosis, hyalinosis of glomerular capillary loops, focal atrophic changes in tubular epithelium were noted. At an acute stage of the disease (up to 10 days), a positive correlation was found between the proliferative index Ki-67 and the activity of ALP and AP phosphatases in the apical parts of nephrothelium, peritubular capillary endothelium ($r_{ALPA}=0.59$; $p=.005$; and $r_{ALPcap}=0.82$; $p=0.000$, respectively (ALP); ($r_{APA}=0.68$; $p=.000$; and $r_{APcap}=0.81$; $p=0.000$, respectively (AP).

Thus, structural renal changes in HFRS complicated by ARF at an acute stage of the disease (up to 10 days) are characterized by a combination of alteration of all nephron elements, which is most pronounced in the tubular part of nephron with damage to the stromal-vascular component, with a concurrent increase in the proliferative activity of convoluted tubular epithelium. Interstitial sclerosis, dystrophic and atrophic changes in the tubules are found in the period after the 21st day of the disease. It has been established that nonspecific phosphatases produce a significant impact on the regeneration processes in renal tubules in HFRS.

Key words: alkaline phosphatase, morphometric, kidneys, proliferative index Ki-67, acute renal failure

● Innate Lymphoid Cells are Activated in Hantavirus Infected Patients and can be Activated by Hantavirus Induced Type I Interferons

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Innate lymphoid cells (ILCs) are involved in immunity and homeostasis but, except for natural killer (NK) cells, their role in human viral infections is not well known. Puumala virus (PUUV) causes hemorrhagic fever with renal syndrome (HFRS), characterized by strong systemic inflammation and immune cell activation. NK cells are highly activated in HFRS, suggesting that also other ILCs might be responding to infection. Here, we phenotypically analyzed peripheral ILCs in PUUV-infected HFRS patients (n=17) and measured plasma levels of soluble factors and viral load. Additionally, we investigated in vitro the mechanisms behind hantavirus-mediated activation of ILCs.

During acute HFRS, the frequencies of NK cells and naïve ILCs were reduced while the frequency of ILC2 was increased. Interestingly, we observed a negative correlation between viral load and frequencies of NK and non-NK ILCs in acute HFRS. Phenotypically, all ILCs showed an activated and proliferating profile. Further, plasma levels of inflammatory proteins, including the ILC-associated cytokines interleukin (IL)-13, IL-23, IL-25, IL-33, and thymic stromal lymphopoietin (TSLP), were elevated during acute HFRS. Moreover, ex vivo sorted ILC2 from uninfected control donors were activated and more viable when exposed to PUUV-infected as compared to uninfected endothelial cells. This activation was dependent on type I interferons (IFNs) produced by infected cells. In line with this, we observed increased plasma IFN- α levels in the acute phase of HFRS.

These findings indicate a general involvement of ILCs in response to human hantavirus infection, and that type I IFNs -produced in response to virus infection- can activate ILCs.

Key Words; innate lymphoid cells, natural killer cells, Puumala virus, hemorrhagic fever with renal syndrome, type I interferon

- **A study of Puumala virus infection during pregnancy using samples from pregnant women and human trophoblast stem cells**

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Puumala orthohantavirus (PUUV) is a member of the genus Hantavirus (family Bunyaviridae). PUUV is a rodent-borne pathogen, and the virus is transmitted to humans by inhalation of infected rodent excreta. In Sweden, Finland, Norway, Russia, and parts of central Europe, PUUV is endemic in bank voles (*Myodes glareolus*). Humans infected with the PUUV may develop a hemorrhagic fever with renal syndrome (HFRS) also known as nephropathia epidemica (NE) and could be lethal in less than 0.5% of the cases. Previously, some case reports showed that not only PUUV infection but also other hantaviruses can cause severe infection in pregnant women. Particularly, the occurrence of a virus across the placenta barrier can infect the fetus and seriously affect the fetal health status.

Pregnancy is a unique condition where pro-inflammatory and anti-inflammatory mediators harmonize creating a protective milieu for the feto-maternal unit by the development of the placenta. Placenta is a vascular organ with maternal and fetal origins and consists of various cell types and is also known as a barrier to various pathogens.

We have detected PUUV RNA in placental tissue collected from a woman who developed symptoms of NE in the 41st week of pregnancy. The viral RNA was detected in both maternal and fetal side of placenta and which indicates transplacental transmission of PUUV. However, the effects of PUUV hantavirus infection during pregnancy have not been well studied. To understand the viral pathogenesis during pregnancy, we developed the novel infection model system by using human trophoblast stem cells derived from the placenta and tested for 1) the susceptibility of PUUV and other related viruses known to cause complications of pregnancy (such as Rift Valley fever virus), 2) the expression of immune-related cytokine mRNA.

● Humoral Immune Characterization of Sin Nombre Hantavirus Survivors

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New World Hantaviruses, including the most common species in North America, Sin Nombre orthohantavirus (SNV), cause an acute respiratory illness in humans called Hantavirus Cardiopulmonary Syndrome (HCPS). HCPS can rapidly progress to respiratory failure and cardiogenic shock, thus limiting the time available for medical intervention. Due to the lack of FDA approved vaccines and therapeutics the case fatality rate of infected individuals remains at ~40%. In this study, we characterized the humoral immune response in SNV survivors to provide a benchmark for optimized vaccine and therapeutic development. We measured antibody effector functions, including neutralization, antibody dependent cellular cytotoxicity (ADCC), and complement dependent cytotoxicity (CDC), using human SNV patient samples ranging from 0 days to 24 years post-hospitalization. The resulting data demonstrates that neutralizing antibody titers remain elevated even 15 years post-hospitalization, with median IC50 values at dilutions just over 1:20,000. Previous studies have demonstrated a positive correlation between high hantavirus neutralizing antibody titers and mild disease outcomes. Our ADCC data suggests that convalescent SNV patients have the potential to develop potent ADCC activity over time. Preliminary data from the CDC assays reveal that complement activity is present mainly in convalescent, versus acute, patients. Collectively, patients who have survived SNV infection have both robust, long lasting SNV specific neutralizing antibodies, and have antibodies with ADCC and CDC effector functions. Our next steps for this study will be to finalize profiling the humoral immune response in SNV survivors and compare these results with the responses from fatal cases of HCPS.

Keywords: sin nombre, hantavirus cardiopulmonary syndrome, convalescence, complement, humoral immunity

● Dobrava and Puumala orthohantaviruses replicate differently in human astrocytes

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In Europe, orthohantaviruses Puumala (PUUV) and Dobrava (DOBV) cause hemorrhagic fever with renal syndrome (HFRS). During the course of HFRS mild neurologic symptoms such as headache, vertigo, and nausea are common. However, severe, potentially life-threatening neurological manifestations are rare and underlying processes are unexplored. Astrocytes, the most abundant glial cells in the central nervous system (CNS) contribute to brain homeostasis in several ways and are also crucial regulators of innate and adaptive immune responses in the CNS.

To investigate the neurotrophic affinity of DOBV and PUUV, we stimulated human astrocytes and neurons in vitro with both viruses and analyzed viral growth dynamics and secretion of selected cytokines and chemokines. We found that both viruses infected astrocytes and neither replicated in neurons. Infected astrocytes showed high secretion of IL-6 and IL-8 after stimulation with both viruses. However, the dynamics of virus growth were different: PUUV started to replicate as early as 2 d.p.i., whereas infectious DOBV progeny were detected as early as 4 d.p.i., but the final titer at 13 d.p.i. for DOBV was 4 log higher than that of PUUV. These results were mirrored also by the percentage of infected astrocytes. Astrocytes infected with PUUV, but not with DOBV, secreted IFN- λ , IP-10 and RANTES, which are associated with CNS infections and play a role in inhibiting viral replication. Thus, it is possible that DOBV inhibits activation of the immune response in astrocytes, which allows its unlimited proliferation and eventually leads to a more severe form of the disease.

Key Words; Dobrava, Puumala, astrocytes, neurons, cytokines

● Defective hantavirus genomes – friend of foe?

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RNA viruses produce pathogen-associated molecular patterns (PAMPs) that can be detected by cellular pattern recognition receptors (PRRs), triggering a signalling cascade that results in the secretion of pro-inflammatory cytokines such as type I/III interferon (IFN) that induces the transcriptional activation of hundreds of antiviral genes. Analysis of the host response in human lung epithelial cells to both pathogenic and apathogenic hantaviruses revealed that all viruses induce a robust and comparable IFN response at the translational and transcriptional level. Investigation of the potential PRRs involved in detecting hantavirus infections clearly demonstrated a role for both MDA5 and RIG-I in triggering the IFN response, while highlighting critical differences in their respective abilities to activate antiviral signalling and restrict viral replication.

Given these similarities in host response *in vitro* across different hantaviruses, how then can we explain widely different degrees in disease severity? In other negative-sense RNA viruses (e.g. IAV and RSV) the accumulation of defective viral genomes (DVGs) have been shown to be associated with highly inflammatory and pathogenic disease. We characterize the first DVGs known for hantaviruses and find evidence for a range of DVGs for both apathogenic and pathogenic hantaviruses *in vitro* and *in vivo*. In particular, an abundant and highly specific DVG produced from the ANDV L segment generates an open-reading frame that could code for a putative novel viral protein. This raises the question whether DVGs, in addition to activating the host immune response, also are able to extend the limited coding capacity of the hantavirus genome.

Key Words: interferon response, viral RNA, defective viral genome, pattern recognition receptor

● Primary human monocytes differentiate into M2-like macrophages upon long-term orthohantaviruses infection

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Primary human monocytes can be permissive or resisting to viral infection and replication. The aim was to examine the elements of the innate immune response of human primary monocytes in vitro infected with pathogenic Puumala orthohantavirus or apathogenic Tula orthohantavirus.

The results have shown that pathogenic as well as apathogenic orthohantaviruses stimulate monocytes' innate immune response, occurring within the first hour after infection with changes at the levels of gene expression, surface molecules/receptors and the synthesis of soluble biological response modulators. Differential immune response was detected upon pathogenic versus apathogenic orthohantavirus infection. Primary monocytes immune response activation after orthohantavirus entry and infection was independent of viral replication. In the early phase of infection, in the interaction with orthohantaviruses, monocytes responded with pro-inflammatory pheno- and geno-type, while in the late phase of infection change of their functional phenotype prevails with regulatory pheno- and geno-type and differentiation towards M2-like monocyte-derived macrophages. Individual donor immune response in gene expression and excretion of soluble response modulators was documented.

Our study identified primary monocytes as transport system for orthohantavirus dissemination in the organism.

Keywords; Puumala orthohantavirus, Tula orthohantavirus, hemorrhagic fever with renal syndrome, innate immunity

● Orthohantavirus reprogramming of transcriptional cellular response in primary human monocytes

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As a front-line defenders against viral invasion monocytes are providing the first virus-cell contact upon infection. Our previous studies showed that monocytes/macrophages are permissive for orthohantaviruses allowing them limited replication in time and amount. Within this study we wanted further to investigate virus to cell interactions focusing on the host cell response and the transcriptional reprogramming of infected cells induced by orthohantaviruses. Primary human monocytes were infected either by Puumala (PUUV) or Tula (TULV) orthohantavirus and cultivated up to seven days. Comprehensive transcriptomic analysis was performed by RNASeq in specific time points post infection.

The results revealed a great number of differentially expressed genes 1h post infection in both PUUV and TULV infection compared to noninfected cells. Most genes were upregulated upon viral infection especially genes involved in cell death. 48h and 7 days post infection the results demonstrated virus induced shift of transcriptional activity compared to early phase with dominance of gene downregulation due to both PUUV and TULV infection. In early hours we found no significant differences in cellular response between the viruses but in later time points PUUV triggered upregulation of genes involved in response to type I interferon compared to TULV.

Key Words; orthohantaviruses, primary monocytes, macrophages, innate immune response, transcriptome

● Expansion of low-density granulocytes by Puumala orthohantavirus infection

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The disease caused by Puumala Orthohantavirus (PUUV) is characterized by strong neutrophil activation. Neutrophils are the most abundant immune cell type in the circulation and are specially equipped to rapidly respond to infections, as has been observed also in the case of COVID-19 pandemic caused by SARS-CoV-2. As part of the massive neutrophil response to SARS-CoV-2, we and others recently detected elevated levels of low-density granulocytes (LDGs) in the circulation of severe COVID-19 patients. In the current study we observed that circulating LDGs are elevated also during acute Puumala orthohantavirus infection. Furthermore, cell surface marker expression analysis revealed that PUUV-associated LDGs are primarily immature and most likely reflects increased neutrophil production from the bone marrow. However, in contrast to SARS-CoV-2 associated LDGs, the secretome of PUUV LDGs did not show significant immunosuppressive ability, which suggests inherent biological differences in the LDG responses that is dependent on the causative virus.

Key Words; Puumala, granulocytes, neutrophils, immature, SARS-CoV-2

Session4. Vaccines, Therapeutics & Clinical aspects

- **Development of a novel minigenome and recombinant VSV expressing Seoul hantavirus glycoprotein-based assays to identify anti-hantavirus therapeutics**

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Seoul virus (SEOV) is a serious health concern because it can cause hemorrhagic fever with renal syndrome (HFRS), which has case fatality rates of ~2%. Currently, there are no approved treatments available for SEOV infections. We developed a cell-based assay system to identify compounds with potential activity against SEOV and characterized their mode of action, and tested several compounds previously reported to have activity against other negative-strand RNA viruses using a newly developed hantavirus antiviral screening systems. We generated a recombinant reporter vesicular stomatitis virus expressing SEOV glycoproteins to confirm whether candidate antivirals target glycoprotein-mediated entry and developed a SEOV minigenome system to identify compounds targeting viral transcription/replication. These systems can be used under lower biocontainment conditions than those needed for infectious viruses. Our findings have important implications for the development of anti-hantavirus therapeutics.

Key Words; orthohantavirus, Seoul virus, Seoul virus minigenome, antiviral, hemorrhagic fever

● Use of Ribavirin during 2018-2019 Andes Sur Outbreak in Chubut, Argentina.

Enzo Lavarra

Context

An outbreak of person-to-person Andes Sur hantavirus disease occurred between October 2018 and February 2019 in an endemic area of Patagonia Argentina, the Andes mountain region in the province of Chubut. A total of 34 cases was reported, 30 of them were admitted in the Esquel Hospital.

Objective

Evaluate the impact of ribavirin-treatment during the early stages of disease on the development of cardiopulmonary syndrome and death.

Methods

A before and after study was used. We compared initial adult cases during the outbreak that not received ribavirin-treatment with those who were enrolled in a ribavirin-treatment clinical assay at the end of the outbreak that enrolled cases during prodromic phase of the disease.

We evaluated the requirement of intensive care unit admission because cardiopulmonary syndrome development and death attributed to hantavirus.

Results

Twenty-one of 25 cases (84%) of before group required for intensive care unit admission, and 10 (40%) died. None of the 4 cases (0%) of after group required for intensive care unit admission neither of them died.

Conclusions

The use of ribavirin during early Andes Sur hantavirus may be useful as treatment of the disease. More investigation is required.

● Clinical and Immunological Biomarkers of Hemorrhagic Fever with Renal Syndrome Outcome during The Early Disease Progression

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The prediction of biomarkers related to disease progression and severity plays a critical role in the prognosis of patients with hemorrhagic fever with renal syndrome (HFRS). However, the knowledge of early prognostic biomarkers of clinical and immunological parameters in HFRS remains unclear. This study aims to evaluate the serum levels of immune function-related proteins and characterize novel biomarkers that may help ascertain the clinical outcomes of HFRS. Enzyme-linked immunosorbent assay, Luminex, and bioanalyzer assays were performed to quantify the 15 biomarkers related to immune function in 49 serum specimens from 26 patients with HFRS. High hemoglobin (HGB) and low urine output (UO) levels were confirmed as potential biomarkers associated with acute HFRS. The serum soluble urokinase plasminogen activator receptor (suPAR) and C-X-C motif chemokine ligand 10 (CXCL10) values were upregulated at the early phase of the diseases. Elevated suPAR, interleukin-10 (IL-10), CXCL10, and decreased transforming growth factor-beta 3 (TGF- β 3) were potential predictors of the disease severity. The overexpression of HGB demonstrated a significant correlation with high levels of suPAR and CXCL10. The downregulated UO positively correlated with increased suPAR, CXCL10, and TGF- β 2, and decreased vascular endothelial growth factor and TGF- β 3. Our results raise significant awareness for physicians regarding prospective biomarkers for monitoring early warning signs of HFRS. These findings provide important insights into the clinical and immunological predictors for disease progression and severity in patients with HFRS to decrease morbidity and mortality.

Keywords; hemorrhagic fever with renal syndrome, biomarker, early phase, severity, prognosis

● New Approaches for the HFRS-Puumala Vaccine Development

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We report some prospects for improving the technology for inactivated PUUV vaccines. To optimize chromatographic purification of the virus, the following sorbents were tested: Capto™ Core 400 and 700 (CC700), Sepharose 4 and 6 Fast Flow (S4FF and S6FF) (GE Healthcare), WorkBeads: WB 40/100, 40/1000, 40/10000 (Bio-Works). CC700 yielded a virus fraction (virus titer 5.1 ± 0.3 lg FFU/ml, total protein <20 µg/ml) comparable in purity to CsCl density gradient ultracentrifugation, with virus loss as high as 97%. Gel filtration at WB 40/10000 allowed obtaining target fractions with a virus titer of 5.3 ± 0.3 lg FFU/ml, a total protein content of ≈ 25 µg/ml, and a virus loss of approximately 90%. The best results were obtained with S4FF and S6FF gel filtration: the loss of target components was about 30%, the virus titer was 4.4 ± 0.3 lg FFU/ml, and the total protein content was ≈ 59 µg/ml.

To find the optimal PUUV inactivation procedure, formaldehyde, β-propiolactone, hydrogen peroxide, UV radiation, and gamma radiation (¹³⁷Cs) were investigated, which differ in the type of interaction with the virus. PUUV purified with CC700 was used for these experiments.

As it turned out, the inactivation method had no statistically significant effect on the neutralizing antibody titer (nAb). Two weeks after double immunization (two weeks apart) with the above virus inactivation methods, the nAb titers in blood sera were: 8.5 ± 0.3 log₂ 6.7 ± 0.4 log₂ in BALB/c mice and Syrian hamsters, respectively. ¹³⁷Cs could be a promising route for virus inactivation because it is producible and, in particular, no residual inactivator remains.

Keywords: PUUV-vaccine, chromatographic purification, virus inactivation methods, immunogenicity, neutralizing antibodies

● Influence of the Immunization Schedule on the Immune Response to Experimental Inactivated PUUV-Vaccine

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More than 97% of HFRS cases in Russia are due to PUUV, therefore this virus plays the main role in the HFRS morbidity structure in Russia. The aim of this study was to determine the optimal immunization schedule with an experimental PUUV-based vaccine for nAb induction in guinea pigs.

The pool of PUUV-infected Vero cell fluids was clarified by filtration, then the virus was concentrated by tangential flow filtration, purified by gel filtration and inactivated with beta-propiolactone. The final preparation contained $5.0 \pm 0.4 \times 10^3$ RNA copies/ml. Guinea pigs were immunized intramuscularly with 0.3 ml, blood was collected from the heart every 2 weeks. The nAbs were determined with the FRNT50.

14 days after the first immunization, the nAbs titer was $5.5 \pm 0.3 \log_2$. After the second immunization at two-week interval, the highest nAbs value was observed on day 56 ($9 \pm 0.2 \log_2$), with subsequent decrease: on day 70 - $7.4 \pm 0.2 \log_2$, on day 504 - $3.9 \pm 0.2 \log_2$.

After two immunizations one month interval, the maximum nAbs titer was observed on day 70 ($9.5 \pm 0.3 \log_2$). Triple immunization at two-week intervals was associated with a similar increase in nAbs as the previous schedule.

Booster immunization 6 and 12 months after double immunization at two-week interval resulted in an increase in nAbs to $10 \pm 0.3 \log_2$ at day 238 and $10.6 \pm 0.2 \log_2$ at day 434, respectively. At the end of the observation period (504 days), nAbs decreased in these groups but was significantly higher than without booster. The optimal immunization schedule is a double injection at 1 month intervals and a booster after 6-12 months.

Keywords: hantavirus, hemorrhagic fever with renal syndrome, PUUV-vaccine, immunogenicity, neutralizing antibodies, immune response, immunization schedule

● Immunomodulatory Effect of Adjuvants in Hantavirus Vaccine in Guinea Pig Model

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The use of adjuvants is of interest to enhance the immune response of the experimental vaccine based on Puumala virus (VAC) or to reduce the dose. The adjuvants selected were: a mixture of short and long O-polysaccharide domains Ac3-S-LPS *Shigella sonnei* (VAC-LPS-1/1) and long-chain (VAC-LPS) at a concentration of 50 mcg/dose; squalene emulsion of the "oil-in-water" type SEPIVACTMSWE (VAC-SWE) - 100 mcg/dose, which had the greatest immunoadjuvant effect in BALB/c mice. VAC with $4.5 \pm 0.5 \lg$ FFU/ml and $2.1 \pm 0.4 \times 10^3$ RNA copies/ml, was inactivated with β -propiolactone and administered to guinea pigs according to the following schedule: 2-fold at 2-week intervals and a booster after 6 months of 0.3 ml/intramuscular.

Peak neutralizing antibodies (nAbs) titer for VAC ($9.2 \pm 0.2 \log_2$) and VAC-SWE ($10.2 \pm 0.2 \log_2$) were reached on day 56, for VAC-LPS-1/1 on day 126 ($8.9 \pm 0.2 \log_2$), and for VAC-LPS from day 14 to 126 ($8.3-9.2 \pm 0.2 \log_2$). The nAbs titer gradually decreased until day 364.

Booster of the vaccines on day 182 resulted in similar nAbs growth profiles for VAC, VAC-LPS and VAC-LPS-1/1, except for VAC-SWE, for which the stimulating effect was minimal.

A 10-fold reduction in immunogen concentration resulted in similar dynamics of nAbs increase on day 56 after booster for VAC ($6.51 \pm 0.3 \log_2$) and VAC-SWE ($6.35 \pm 0.3 \log_2$) up to $7.57 \pm 0.2 \log_2$ and $8.23 \pm 0.3 \log_2$ on day 294 and 238, respectively. No immunomodulatory efficacy was observed for all LPS under these conditions.

The use of adjuvants thus leads to a statistically significant prolongation of the immune response, extended the intervals in the immunization schedule for an acceptable immune response.

Keywords: hemorrhagic fever with renal syndrome, PUUV-vaccine, adjuvants, immune response, immunization schedule

● Replication-competent Recombinant Vesicular Stomatitis Viruses Evaluated as Potential Vaccine Candidates Against Old World Hantaviruses

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Recently, Warner, Stein, et al. have demonstrated that replication-competent, recombinant viruses (rVSVs) bearing New World hantavirus glycoproteins can confer protection by vaccination in a lethal New World ANDV and SNV challenge in Syrian hamsters. We aimed to broaden the potential application of this method of vaccination to hantaviruses in the Old World clade. With the help of two point mutations, we have been able to successfully generate rVSVs that display Old World hantavirus Gn/Gc. These viruses can replicate to high titers in tissue culture, making exploratory vaccine studies in rodents amenable. As a prelude to such challenge studies, we evaluated the immunological properties of rVSVs expressing different Old World hantavirus Gn/Gc proteins in mice. Analysis of the immune sera from these animals showed that the rVSVs could elicit neutralizing antibodies against their cognate glycoproteins. We found the duration of immunogenicity to last up to twelve months post vaccination. Moreover, some (but not all) of these rVSVs could also elicit antibodies with cross-neutralizing activity against divergent Old World hantavirus Gn/Gc proteins. Overall, we found a potential candidate similar to those for New World hantaviruses involving rVSV surrogates as vaccine vectors to elicit protection against infection.

Keywords: surrogate system, vaccine candidate

● Cardiac arrhythmia and reduced heart rate variability in patients with Puumala orthohantavirus infection

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Orthohantaviruses are rodent borne viruses causing hemorrhagic fever with renal syndrome and hantavirus cardiopulmonary syndrome depending on the causative virus. Cardiac involvement is frequent and could contribute to the clinical symptoms and fatal outcome in severe cases. There are, however, gaps of knowledge regarding disturbances in rhythm and electric conduction of the heart. Our aim was to investigate cardiac arrhythmia in patients with Puumala hantavirus (PUUV). We used 24-h Holter electrocardiogram (ECG) monitoring which is more sensitive to detect disturbances in electric conduction and rhythm compared to normal ECG. Holter ECG was performed in 21 hospitalized patients at Umeå University Hospital, Sweden (13 women and 8 men, median age 60 years) with verified acute PUUV-infection and at a follow-up after 3 months. The heart rate variability (HRV) data were compared with HRV data from 64 healthy controls. In all, 63% of the patients had at least one abnormal finding detected by standard Holter ECG analysis during the acute phase of the disease. The most common was supra-ventricular arrhythmia (42%), including one patient with atrial fibrillation. Sinus bradycardia was found in six patients (32%) and ventricular arrhythmia found in two patients (11%). Notably, four patients displayed clear evidence of reduced HRV during the acute phase that normalized at follow-up. Patients with orthohantavirus infection display cardiac arrhythmias indicating cardiac involvement in the acute infection. Further, decreased HRV was found in several patients during the acute phase which could indicate disturbances in the autonomic regulation of the heart.

Session5. Diagnosis & Emerging Hantaviruses

● ***Puumala orthohantavirus* diagnostic at the Swedish National Reference Laboratory**

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Puumala orthohantavirus is a hantavirus that is found in northern Sweden, where it causes a mild form of haemorrhagic fever with renal syndrome (HFRS). Symptoms can include fever, headache, gastrointestinal symptoms, and impaired renal function. The mortality rate is less than 1%. Bank voles act as reservoir for *Puumala* viruses, and the disease has subsequently been called "sorkfeber" (Eng. "vole-fever"). Humans get infected through contact with infected rodent material, e.g. inhalation of dried fecal matter.

The Clinical Microbiology Department of University Hospital of Umeå (Norrlands Universitetssjukhus) is the Swedish National Reference Laboratory for Hantaviruses (supported by the Public Health Agency of Sweden) and assists with advice on sampling and choice of analysis. Serological analyses have been the first choice to diagnose infection, supplemented by PCR in cases of patients with antibody deficiency or very early in the course of the disease.

We have recently changed method for the detection of antibodies against *Puumala* virus from in-house immunofluorescence to a commercial ELISA kit. We are also in the process of changing the workflow to deliver faster results to our customers.

Here we present the verification of two commercially available ELISA kits from Reagentia for the detection of *Puumala* IgM and IgG antibodies, as well as the new and improved workflow, which uses PCR as screening and ELISA as confirmatory tests.

Key Words; *Puumala*, serology, ELISA, diagnostics

● Amplicon-based nanopore sequencing and rapid genome surveillance of Hantaan virus from *Apodemus agrarius*, Republic of Korea

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The high-throughput technology for whole-genome sequencing of etiologic agents plays an important role in the detection and characterization of emerging viruses. A nanopore technology-based device is a portable platform that allows real-time sequencing in fields or hospitals. Hantaan virus (HTNV) causes hemorrhagic fever with renal syndrome (HFRS) and poses a major public health threat. We evaluated the feasibility of nanopore system for whole-genome sequencing of HTNV from rodent samples having different viral copy numbers. Multiplex PCR-based nanopore sequencing was performed from lung tissues of *Apodemus agrarius* from the Republic of Korea. The amplicon-based high-throughput sequencing using nanopore platform generated nearly full-length genomic sequences of HTNV and demonstrated sufficient read depth for analysis of phylogenies, within 8 h of sequencing. The mean identity of the HTNV genome sequences for the nanopore sequencer compared to those of generated from Illumina MiSeq were 99.8% (L and M segments) and 99.7% (S segment), respectively. These findings highlight the potential of the portable nanopore sequencer for rapid generation of accurate genomic sequences of HTNV for quicker decision making in point-of-care testing of HFRS patients during a hantavirus outbreak.

Key Words; MinION sequencing, amplicon-based sequencing, whole-genome sequencing, viral genome surveillance, Hantaan virus

● A rapid mobile diagnostic assay of *Seoul orthohantavirus* harbored by *Rattus norvegicus* collected in Gangwon Province, Republic of Korea

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Seoul virus (SEOV) infection occurs worldwide due to the global distribution of natural reservoir hosts (brown rat, *Rattus norvegicus*; black rat, *R. rattus*). SEOV-related hemorrhagic fever with renal syndrome (HFRS) is responsible for about 20% of clinical cases with a mortality rate of <1%. This study aims to evaluate the feasibility of a portable Franklin real-time thermocycler (Biomeme, Philadelphia, PA, USA) for facilitating rapid decision-making of SEOV infection. 27 wild rats (*R. norvegicus*) were captured from Seoul Metropolitan City and Gangwon Province in the Republic of Korea (ROK), during 2016–2020. The serological and molecular positivity for SEOV was 5/27 (18.5%) and 2/27 (7.4%), respectively. SEOV RNA was detected and quantified in multiple tissues of rats by the Biomeme platform, with differences in Ct values ranging from 0.6 to 2.1 cycles compared to a benchtop Quantstudio 5 Flex Real-time PCR system (Applied Biosystems, Foster City, CA, USA). Using amplicon-based next-generation sequencing, whole-genome sequencing for SEOV was performed from lung tissues of Rn18-1 and Rn19-5 collected in Gangwon Province, ROK. The phylogenies of SEOV genomes revealed a phylogeographical diversity of rat-borne orthohantavirus collected in Gangwon Province. Finally, we report a novel isolate of SEOV Rn19-5 from wild *R. norvegicus* collected from Gangwon Province. Our findings showed that the mobile Biomeme device can be applied for the genome-based diagnosis of SEOV comparable to the laboratory-based system. Our findings provide useful insights into the field-deployable diagnostic assays and genetic diversity of orthohantaviruses for the rapid response to hantaviral outbreaks in the ROK.

Key Words; Seoul virus; hemorrhagic fever with renal syndrome, Biomeme system, portable diagnostic assay, molecular diagnosis, next-generation sequencing, whole-genome sequencing, genetic diversity, hantavirus isolation

● VirPipe: An Easy-to-use and Customizable Pipeline for Detecting Hantaviral Genomes from Next-generation Sequencing

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Next-generation sequencing (NGS) has been extensively applied in the identification and discovery of pathogens. However, the number of virus detection pipelines supporting MinION sequencing has been significantly limited. Moreover, existing pipelines have weaknesses that hinder their active use in research, such as discontinued maintenance and difficulties in installation.

Here, we present VirPipe, a novel bioinformatic pipeline for the detection and analysis of viral genomes from Nanopore or Illumina sequencing inputs. The VirPipe implements three distinct analysis methods, which were reference mapping, taxonomic classification, and contig analysis. The reference mapping and taxonomic classification enhance the sensitivity of detecting known viruses. In addition, contig analysis after assembly of reads is used for the analysis of novel virus sequences, including the potential of human-animal transmissibility (zoonosis). This VirPipe leads users to install the pipeline with only a few commands and customize it for targeted and metagenomic approaches using both Nanopore and Illumina platforms.

To examine its utility, we ran VirPipe with published NGS datasets of Hantaan orthohantavirus (HTNV) as inputs. In the reference mapping, all three segments of HTNV were almost entirely covered by the input reads. The taxonomic classification exhibited that a majority of the reads were classified into HTNV. Finally, numerous assembled contigs showed high similarity with HTNV reference sequences by the contig analysis. The VirPipe analysis demonstrated the high capacity for mapping and classification of HTNV-related reads by performing MinION and MiSeq. In conclusion, this pipeline provides a useful utility for virologists to identify and examine a broad range of viral genomes, due to its easy and adjustable properties in handling bioinformatics tools.

Keywords: Viral genomics, Bioinformatic pipeline, Reference mapping, Taxonomic classification, Contig analysis

● On-site Rapid Detection and Monitoring of Hantaan orthohantavirus Using Real-time PCR in Wild Rodents (*Apodemus agrarius*)

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Hantaviruses are present in a variety of natural hosts, including bats, moles, shrews and carnivores as well as rodents. In Eurasia, hemorrhagic fever with renal syndrome (HFRS) caused by old world hantaviruses, and in the Americas, hantavirus pulmonary syndrome (HPS) caused by new hantaviruses are the main causes. In Korea, 300 to 500 patients are diagnosed each year, and both civilian and military patients continue to occur. HFRS is transmitted through the respiratory tract and has a high fatality rate, making it highly likely to be used in a bioterrorist attack. Point-of-care testing (POCT) is a very useful technique because it can quickly identify pathogens with less space constraints and fewer specialized personnel required for testing compared to laboratory diagnostics. In this study, we conducted an efficacy trial of an optimal set of primers and TaqMan probes designed to detect Hantaan orthohantavirus (HTNV) using Biomeme Franklin among point-of-care diagnostic devices. We also assessed the feasibility of POCT for HTNV infection by collecting *Apodemus agrarius*, the host of HTNV, in the field and extracting RNA from whole blood, urine, saliva, and feces using M1 RNA/DNA preparation cartridges.

Keywords: hantaan orthohantavirus, biomeme franklin, point-of-care testing, taqman probe, field

● Development and Evaluation of a rapid molecular diagnostic system as duplex q RT-PCR for differentiation of Hantann/Seoul viruses in single slot

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Hemorrhagic fever renal syndrome (HFRS) is caused by Hantaviruses and has recognized as persistent public health concerns to human in Asia and Europe. The diagnostic method for HFRS is insufficient for suspected patients and it is important to conduct a rapid and accurate diagnosis for proper treatment in time. Five species of Hantaviruses, Hantaan virus (HTNV), Seoul virus (SEOV), Imjin virus, Muju virus, and Soochung virus, were detected in South Korea. Among them, two genotypes of SEOV and HTNV are pathogenic to human infection, whereas rest of three viruses were not elucidated yet.

We developed real-time RT-PCR for five viruses with designed primers/probes targeting to S segment of each virus and then was verified with transcribed viral RNAs. Each primer/probe set presented high sensitivity and specificity. And we established a duplex qRT-PCR for HTNV and SEOV and evaluated functional analytics either by positive control DNA or clinical specimens. It was proven that developed assay is capable for presenting high sensitivity of less than 10 copies of target virus, respectively, and no cross-reactivity to other viruses belongs to *Bunyaviridae*. Then, it was validated with 120 clinical samples that were diagnosed by IFA to HTNV, in resulted of detected positive of HTNV from 5 specimens that were positive by IFA.

In conclusion, we developed two types of molecular diagnostic method; single assay for five species individually and duplex qRT-PCR for HTNV/SEOV. They are useful for rapid and efficiently diagnosis of HFRS and available for differentiating detection of Hantaviruses on survey in human and main natural vectors.

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Key Words; duplex qRT-PCR, Hantann Virus (HTNV), Seoul Virus (SEOV)

AUTHOR INDEX

• Agnidipta Ghosh	41	• Austin Leach	51
• Ai Hayashi	27, 55, 60	• Ayato Takada	60
• Åke Lundkvist	43, 45	• Aydar Ishmukhametov	29, 80, 81, 82
• Akina Mori	60	• Balachandran Manavalan	88
• Akio Koizumi	53	• Barbara Anđelić Dmitrović	65
• Aleksandar Radulovic	61	• Barbara Knust	28
• Alemka Markotić	35, 74, 75	• Barry Rockx	38, 39
• Alexander Plyusnin	43, 45	• Bartłomiej Porebski	42
• Alexandra Balkina	29	• Bartonička T	56
• Alexandra Schlickmann Pereira	63	• Beata Sikorska	55
• Alexandra Serris	23	• Benjamin Nilsson-Payant	73
• Alexandro Guterres	64	• Bernard M Hang'ombe	60
• Alicia M. Moreau	41	• Bernardo Rodrigues Teixeira	63
• Alison M Kell	36	• Bohyeon Kim	48
• Amy Whitesell	28	• Bohyun Kim	88
• Ana Kuehne	41	• Boris Klempa	31
• Anders Blomberg	84	• Borut Furlani	72
• André Felipe Andrade Santos	64	• Brian Amman	28
• Andrew M. Skidmore	58	• Briana Spruill-Harrell	26
• Andrew S. Herbert	23, 41, 44, 83	• C. Garrett Rappazzo	23
• Angela Sloan	73	• Caitlin M. Cossaboom	28, 51
• Anna Carrasco García	69	• Casey C. Perley	37
• Anna Rodén	85	• Catalina Florez	23, 44
• Anna Smed-Sörensen	69	• Cecilia M. O'Brien	23, 44
• Anna Vetrova	80, 81, 82	• César Albariño	77
• Anna Z. Wec	23	• Chandika Gamage	53
• Anouk van Haperen	38	• Cheng-Feng Chiang	28
• Anton Jan van Zonneveld	38	• Choon-Mee Kim	48
• Antonia Paic	65	• Christian Sieben	66
• Antonija Pajić	35	• Christina F. Spiropoulou	77
• Antti Vaheri	40, 43, 69	• Chunyan Ye	23, 30, 71
• Anu Kantele	40	• Cibeles Rodrigues Bonvicino	63
• Ardith Gibbons	28	• Cinthia Avila Ardaya	51
• Åsa Fahlman	52	• Clas Ahlm	23, 70, 76, 84
• Ashwini Kedari	45	• Colleen B. Jonsson	26
• Atanas V. Demirev	62	• Connie S. Schmaljohn	19
• Augustine Natasha	48, 88	• Courtney A. Cohen	83

• Daesang Lee	50, 87, 89	• Fuka Kikuchi	27, 55, 60
• Dafna M. Abelson	23	• Georges Verjans	39
• Danny Noack	38, 39	• Geum-Young Lee	50, 57, 79, 86, 87
• Darrell L. Dinwiddie	49	• Gianina Arata	33
• Daryl B. Domman	49	• Giulia Vanoni	69
• David Safronetz	73	• Göran Bucht	52
• Debi Cannon	28	• Gorana Stamenkovic	61
• Denis Polančec	74	• Gorka Lasso	41
• Denise Haslwanter	23	• Gustavo Palacios	21
• Devinda S. Muthusinghe	53, 60	• Hajime Kamiya	27, 55, 60
• Diana Hanfelt-Goade	49	• Hana Zelená	47
• Domagoj Tomasović	35	• Hayne Cho Park	50, 79
• Dominique André	85	• Heekyung Cho	86
• Dong Hyun Song	50, 87, 89	• Hee-Young Lim	90
• Dong-Min Kim	48	• Heung-Chul Kim	86
• Eduardo A. Bignon	33	• Hiroaki Kariwa	67
• Ekaterina M. Luchnikova	59	• Hyeongseok Yun	87, 89
• Elba Regina Sampaio de Lemos	63, 64	• Ilona Rissanen	45
• Eleanor Marshall	38	• Il-Ung Hwang	87
• Eliisa Kekäläinen	40	• Inna Krapivunaya	28
• Elizabeth Shedroff	28, 51	• Ivana Pavičić	75
• Emir Talundzic	28	• Ivan-Christian Kurolt	65, 74, 75
• Emma S. Esterman	23	• Jaehun Jung	50, 79
• Emmanuel Kodua	60	• Jairo Mendez-Rico	51
• Emmanuel Messias Vilar	64	• James C. Geoghegan	23
• Enzo Lavarra	78	• Janne Tynell	23, 46
• Esteban Rodríguez	33	• Janusz Hejduk	55
• Eun-Mee Park	90	• Janusz Markowski	55
• Eunyoung Yang	86	• Jay W. Hooper	25, 37, 44
• Eva Mittler	23, 34, 46, 83	• Jeewan Choi	87
• Evans Nnamani	26	• Jenny Mjösberg	69
• Evgeniy Tkachenko	29, 80, 81, 82	• Jeong-Ah Kim	50
• Evseyev AN	68	• Jernej Jorgačevski	72
• Ezgi Kasikci	34	• Jerry W. Dragoo	49
• Fadi Alnaji	73	• Jieun Park	48, 88
• Felix A. Rey	23, 33	• Jimmy Revollo Guzmán	51
• Fernando de Oliveira Santos	63	• Jin Il Kim	62
• Filip Rokić	75	• Jin Sun No	50, 57, 79, 86
• Florian Krammer	39	• Jinyeong Lee	50, 57, 79, 86, 87
• Fořtová A	56	• Jin-Won Song	48, 50, 57, 62, 79, 86, 87, 88, 89

• Jiro Arikawa	53	• Kimia Maleki	69
• Joel M. Montgomery	28, 51, 77	• Kkothanahreum Park	86
• Johan Rasmuson	84	• Kouji H. Harada	53
• Johanna Tauriainen	69	• Kristian Bodulić	75
• Johanna Tietäväinen	40	• Kumiko Yoshimatsu	53, 60
• John D. Klena	28, 51	• Kurt C. Schwalm	49
• John Dye	44	• Kyungmin Park	48, 50, 57, 79, 86, 87, 88, 89
• John Klena	51	• Kyuyoung Lee	62
• John M. Dye	23, 41, 44, 83	• Lana Jakopec	35
• Jonas Klingström	24, 42, 69	• Larry Zeitlin	23
• Jonas Näslund	52	• Laura C. Polanco	23
• Jonathan L. Dunnum	49	• Laura M. Walker	23
• Jongwoo Kim	48, 50, 57, 79, 86, 87, 88	• Laura Menke	66
• Joo-Yeon Lee	90	• Leanna Sayyad	51
• Jordan J. Clark	39	• Leszek Rychlik	55
• Jorlan Fernandes	63, 64	• Lev Levanov	43, 45
• Joseph A. Cook	49	• Lidija Cvetko Krajinović	74, 75
• Judith van den Brand	39	• Linda Bjedov	65
• Jukka Mustonen	40, 43, 69	• Lindsey Estetter	28
• Julia Brignone	51	• Lionel Gresh	51
• Julu Bhatnagar	28	• Liudmila Yashina	59
• Jung-Eun Kim	87, 89	• Lorenz Wirth	69
• Jungsang Ryou	90	• Lothar Jänsch	66
• Jussi Hepojoki	40, 43	• Lucia M. Principe	37
• Juwon Park	87	• Luz Cabrera	40, 76
• Juyoung Noh	86, 87	• Lwande Olivia Wesula	52
• Karen Yamada	27	• M.V.M.L. Jayasundara	52
• Karla Svaguša	35	• Magnus Evander	70
• Kartik Chandran	23, 30, 34, 41, 44, 46, 49, 83	• Maja Potokar	72
• Katariina Vapalahti	43	• Man-Seong Park	62
• Katarina Resman Rus	72	• Marcin Markowski	55
• Kathryn E. Coan	49	• Marco Goeijenbier	38
• Kazuma Tamiya	67	• Marco van Ham	66
• Keleş S.J.	54	• Mari Kanerva	76
• Ken Maeda	27	• Maria Alejandra Morales	51
• Kenta Shimizu	53	• Maria E Morales-Betoulle	28, 51
• Ketan Patel	51	• Maria Laura Martin	51
• Keun Hwa Lee	57	• Maria Morales-Betoulle	28
• Kijin Kim	88	• Marília Lavocat Nunes	63
• Kimberly Martinez	36	• Marina García	69

• Marina Siljic	61	• Rebecca L. Brocato	37
• Marion P.G. Koopmans	38	• Rebekah M. James	83
• Mariya Balovneva	29, 80, 81, 82	• Remigius Gröning	23, 46
• Mariya Egorova	29, 80, 81, 82	• Renata Carvalho de Oliveira	64
• Markus Keller	23	• Richard Yanagihara	27, 55
• Mary Choi	28	• Robert A. Nofchissey	30, 49, 58
• Masahiro Kajihara	60	• Robert D. Owen	26
• Masashi Harada	27	• Robert K. Kim	37
• Mattias Forsell	46, 76	• Robert Zorec	72
• Mattias N.E. Forsell	23, 46, 76	• Rohit K. Jangra	23, 34, 41, 83
• Megan M. Slough	23, 34, 41, 83	• Rola J.	54
• Melanie Rissmann	39	• Rolf Hörnsten	84
• Melissa Mobley	28	• Rommel Paneth Iheozor-Ejiofor	43, 45
• Michael R. Wiley	37	• Rong Li	41
• Miša Korva	20, 72	• Rosanne W. Koutstaal	38
• Mojgan Deldari	28	• Rostislav Teodorovich	29, 80, 81, 82
• Motoi Suzuki	55	• Roxana Loayza Mafayle	51
• Müller T.	54	• Russ Bakken	44
• N.P. Sunil-Chandra	52	• Russell R. Bakken	23, 41
• Na Yeon Jang	48, 88	• Růžek D	56
• Nelly Mendoza Loayza	51	• Samuel M. Goodfellow	30, 49, 58
• Nicole D. Tischler	33	• Sangyi Lee	62
• Oksana Leonovich	29	• Sanna Mäki	40
• Olli Vapalahti	40, 43, 45, 76	• Sara Puspareni Prayitno	48, 88
• Orłowska A.	54	• Sara Zufan	28
• Oscar Fernandez-Capetillo	42	• Satoru Arai	27, 55, 60
• Pablo Guardado-Calvo	23, 33	• Satoshi D. Ohdachi	27
• Patrícia Dias Galvão Alves	64	• Satu Mäkelä	40, 43, 69
• Patricija Pozvek	72	• Se Hun Gu	50, 87, 89
• Patrick Jesus de Souza	64	• Sejik Park	62
• Paulo Sérgio D'Andrea	63	• Seong Tae Jeong	50
• Pavani Senarathne	53	• Seong-Gyu Kim	86, 87
• Paweł P. Liberski	55	• Seonghyeon Lee	48, 88
• Payel Chatterjee	77	• Sergey A. Abramov	59
• Pedro Cordeiro Estrela	64	• Seung Ho Lee	86
• Peter van Run	39	• Seungchan Cho	50, 57, 79, 86, 87
• Petr Tkachenko	29	• Seung-Ho Lee	50, 57, 79, 86, 87, 89
• Petra Svoboda	65, 74, 75	• Shaanika Nanayakkara	53
• Punya Shrivastava-Ranjan	77	• Shailesh Budhathoki	86
• Rainer G. Ulrich	23, 46, 66	• Shannon L.M. Whitmer	28, 51

• Sharon Messenger	28	• Thayssa Alves Coelho	63, 64
• Shigeru Morikawa	27	• Thomas Spliethof	39
• Shilpi Jain	77	• Tomas Strandin	23, 40, 43, 46, 69, 76
• Shintaro Kobayashi	67	• Tonilynn Baranowski	71
• Shivani Rajoriya	48, 88	• Trevor Shoemaker	51
• Shuzo Urata	53	• Urban Wiklund	84
• Silmara Aparecida Ferreira de Carvalho	63	• Valentina Cirkovic	61
• Simo Miettinen	40	• Valerie J. Morley	49
• Simone Sidoli	23	• Vasyi Khodzinskyi	55
• Sithumini M.W. Lokupathirage	53, 60	• Vincent van Duinen	38
• Sofia Lundström	70	• Wanda Christ	42
• Stefan Klimaj	36	• Whitney Weigel	69
• Stella Mrmić	65	• Won-Keun Kim	48, 50, 57, 79, 86, 87, 88
• Stephanie R. Monticelli	41	• Xiankun Zeng	37, 41
• Steven B. Bradfute	23, 30, 49, 58, 71	• Yan Sun	23
• Steven C. Almo	41	• Yanan Liu	41, 44
• Sun-Hee Jung	90	• Ye-Ji Lee	90
• Suvi Jokiranta	40	• Yong-Dae Gwon	70
• Svetlana Kurashova	29, 80, 81, 82	• Yongjin Qiu	60
• Tae-Hun Lim	48, 48	• Yongjin Yi	50, 79
• Tamara Dzagurova	29, 80, 81, 82	• You Mi Lee	48
• Tarja Sironen	43	• Young-Su Kim	86
• Tatiana Pádua Tavares de Freitas	63	• Yu-Jin Kim	86
• Tatiana V. Tregubchak	59	• Yulia Popova	29, 81
• Tatjana Avšič Županc	72	• Željka Mačak Šafranko	35
• Teemu Smura	46	• Zhongde Wang	41, 44
• Terry A. Klein	86	• Zrinka Štritof Majetić	65
• Tetsuya Mizutani	55	• Zuoxing Wei	60

GCFLU Quadrivalent is an egg-based seasonal influenza vaccine developed and manufactured by GC Biopharma, which is a WHO PQ vaccine.⁵

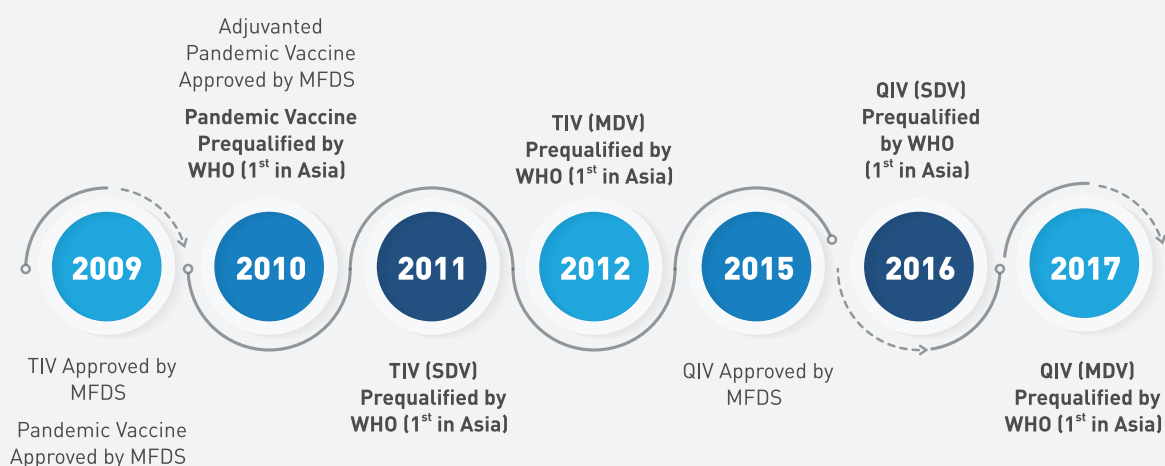
Product Profile



Type	Description
Vaccine Type	Inactivated Quadrivalent Seasonal Influenza Vaccine
Vaccine Strain	WHO Recommended Strains
Virus Culture	Fertilized Egg
Administration	0.5mL Intramuscular Injection
Age Indication	Persons Aged 6 Months and Older
Container	Pre-filled Syringe
Preservative	Thimerosal Free
Manufacturer	GC Biopharma

History

GCFLU TIV and GCFLU QIV have obtained the **WHO prequalification**.⁶



WHO Prequalified Seasonal Influenza Vaccines⁶

Prequalified	Type	Commercial Name	Pharmaceutical Form	Presentation	No. of Doses	Manufacturer
12/04/2011	Influenza, seasonal (Trivalent)	GCFLU inj.	Liquid: ready to use	Vial	1	GC Biopharma
07/11/2012	Influenza, seasonal (Trivalent)	GCFLU Multi inj.	Liquid: ready to use	Vial	10	GC Biopharma
21/12/2016	Influenza, seasonal (Quadrivalent)	GCFLU Quadrivalent inj.	Liquid: ready to use	Vial	1	GC Biopharma
03/04/2017	Influenza, seasonal (Quadrivalent)	GCFLU Quadrivalent Multi inj.	Liquid: ready to use	Vial	10	GC Biopharma

- WHO prequalification ensures vaccines used in immunization programmes are safe and effective.⁷
- It provides Member States and procurement agencies, such as UNICEF.⁷
- Once a vaccine is prequalified and introduced to the market, WHO ensures it continues to meet standards.⁷

MDV, multi-dose vial; PQ, pre-qualification; SDV, single-dose vial; TIV, trivalent influenza vaccine; QIV, quadrivalent influenza vaccine; WHO, World Health Organization; MFDS, Ministry of Food and Drug Safety.